

AI Exposure and Entrepreneurship*

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Abstract

This paper asks whether artificial intelligence is changing the prevalence of growth-oriented entrepreneurship. In a commuting-zone-by-sector panel of U.S. workers, we compare sectors with differing AI exposure within the same commuting zone and year, absorbing every shock common to a place. A one-standard-deviation increase in AI exposure lowers the incorporated self-employment share by 0.401 percentage points (10.8% of baseline) in 2019-2024, while unincorporated self-employment shows no response. The decline first emerges in 2019, four years after the deep-learning era began, and is concentrated in the managerial and professional occupations from which incorporated business owners are drawn.

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1 Introduction

Young firms create a disproportionate share of American jobs. Decker et al. (2014) show that startups account for about 20 percent of gross job creation, and high-growth firms, which are disproportionately young, account for almost half. Although most entrants fail or stay small, the lasting contribution to job creation comes from the few survivors that eventually grow. Because these firms are built by the entrepreneurs who found and own them, sustaining the flow of such founders is a policy-relevant matter.

General-purpose technologies in particular have historically diffused not only through the firms that invent them, but through waves of new entrants that adapt them to new products, markets, and industries: electrification, the semiconductor, and the commercial internet each generated such waves (Bresnahan and Trajtenberg, 1995; Gompers, Lerner, and Scharfstein, 2005). In this paper, we treat artificial intelligence¹ as a general-purpose technology coming from the deep-learning systems that firms began adopting around 2015, of which large language models are the latest generation. We ask whether this technology increases growth-oriented entrepreneurship, as earlier general-purpose technologies did, or reduces it. Do the sectors most exposed to AI have more growth-oriented entrepreneurs, or fewer, while their subsistence margin remains unchanged?

The empirical literature on AI and labor markets measures how AI exposure affects hiring, wage growth, and task reallocation within exposed occupations (Acemoglu, Autor, et al., 2022; Webb, 2020; Hampole et al., 2025; Johnston and Makridis, 2026; Humlum and Vestergaard, 2025). Whether exposed labor markets continue to produce entrepreneurs has drawn little attention (Fossen and Sorgner, 2021). The gap matters beyond the AI literature. Business entry in the United States has declined for four decades, and since 2000 this decline has been accompanied by a shrinking role for high-growth young firms (Decker et al., 2014; Decker et al., 2016). Innovation policy, meanwhile, leans on the assumption that frontier technologies reach the broad economy through startups and subsidizes that channel accordingly. Whether AI increases or reduces the prevalence of growth-oriented entrepreneurship is therefore an empirical question with policy consequences.

We create a commuting-zone-by-sector-level measure of occupational AI exposure, built by weighting the Eloundou et al. (2024) occupation-level exposure scores over each cell’s pre-period occupation mix. Eloundou et al. (2024) rate, task by task, whether direct access to a large language model would materially accelerate that task, and aggregate the ratings to

¹Throughout the paper, “AI” denotes the modern wave of systems built primarily on deep-learning methods, including large language models; strictly speaking, deep learning is a subset of the broader field of artificial intelligence.

an exposure share for each occupation; we assign that share to a commuting-zone-by-sector cell according to the occupations employed there before the deep-learning era began. The measure therefore varies not because a place specializes in a sector with higher AI exposure, but because the same sector is staffed by a different mix of occupations from one local labor market to the next, which is what lets us compare exposure within a commuting zone rather than across places.

We measure growth-oriented entrepreneurship as incorporated self-employment, the margin Levine and Rubinstein (2017) show is held by owners who hire employees, hold capital, and earn returns to cognitive skill, as distinct from unincorporated own-account work closer to subsistence and gig activity. We build a panel of 741 commuting zones by 15 sectors over 2010–2024 from the American Community Survey (165,495 cells) and assign each commuting-zone-by-sector cell its AI exposure. On this panel we estimate event studies saturated with commuting-zone-by-sector, commuting-zone-by-year, and sector-by-year fixed effects, so that identification comes from sectors that differ in occupational exposure inside the same commuting zone in the same year. The commuting-zone-by-year fixed effects are the core of this design: they absorb every shock to a place in a year, including the local college share and its trend, agglomeration economies, housing costs, credit conditions, and pandemic-era demand, so that no characteristic of high-AI-exposure places can drive the estimates.

Because the ACS records self-employment as of the survey date, our outcome is a stock share of the employed population rather than a flow of business formations or closures. The design identifies whether AI exposure changes the prevalence of incorporated ownership; on its own it cannot attribute that change to slower entry, faster exit, or some combination of the two, a limitation we return to in the conclusion.

We find that growth-oriented entrepreneurship, measured as the share of workers who are incorporated business owners, declines with occupational exposure to AI. The incorporated self-employment rate falls by 0.144 percentage points per standard deviation of exposure in 2015–2018 (significant at the 5 percent level) and by 0.401 percentage points in 2019–2024 (significant at the 1 percent level), a 10.8 percent decline against the 3.71 percent baseline rate; pre-2015 coefficients are jointly indistinguishable from zero. The decline is concentrated on entrepreneurs who build firms: the unincorporated, own-account margin does not respond under our primary measure (−0.078 percentage points, insignificant), absorbing at most a third of the incorporated margin’s proportional decline, a pattern inconsistent with a shock to self-employment broadly.

The occupations in which incorporated owners are concentrated explain this pattern: an occupation’s incorporated self-employment rate rises with its AI exposure (+0.20 correlation)

while its unincorporated rate falls (-0.17), and the top exposure tercile accounts for 56.0 percent of all incorporated self-employment on 37.7 percent of employment. AI exposure therefore reaches incorporated ownership through the managerial and professional work that AI-exposed workers do, not only through their sector’s product market.

The decline emerges in 2019, four years after the 2015 onset of the deep-learning era, consistent with the pace at which firm-level AI investment scaled over 2016–2018 (Babina et al., 2024). It precedes both consumer language-model products and the pandemic by several years, and persists through the 2021–2024 business-formation boom, when the commuting-zone-by-year fixed effects absorb any local pandemic severity.

The evidence is consistent with AI complementing the scale, data, and organizational capital that incumbents hold, so employment inside incumbents becomes more attractive relative to incorporated business ownership. Babina et al. (2024) document the corresponding firm-level pattern, AI investment concentrating in the largest firms; our result is its labor-market counterpart, formalized in Appendix A.

These findings are robust across exposure measures, samples, aggregations, and data sources: the Felten, Raj, and Seamans (2021) measure, built independently of language models, correlates 0.918 with ours and delivers a similar estimate (-0.443), and IRS-based administrative records corroborate the survey-based decline. The most plausible alternative explanation, the 2017 tax reform’s Section 199A exclusion, does not survive a direct test against the AI measure: controlling for it leaves the AI coefficient unmoved (-0.407 against -0.401), and a structural-break test places only 4 percent probability on 2018, the year a tax-driven reorganization would require.

The paper contributes to several strands of literature. First, it contributes to the empirical literature on AI and labor markets, which measures what the technology does to employment, wages, and the task content of jobs (Acemoglu, Autor, et al., 2022; Acemoglu and Restrepo, 2020; Acemoglu and Restrepo, 2022; Brynjolfsson, Mitchell, and Rock, 2018; Deming, 2017; Humlum and Vestergaard, 2025; Johnston and Makridis, 2026). This literature treats workers as employees whose jobs are reshaped, displaced, or complemented by the technology; it has not asked whether the same exposure that reshapes a job also changes whether the person holding it owns an incorporated business. We add the entrepreneurship margin, estimated within local labor markets so that place-level confounds are ruled out by design, following the commuting-zone exposure design of Acemoglu and Restrepo (2020).

Second, it contributes to the literature on declining business dynamism, which documents a decades-long fall in high-growth entry in the United States and has not settled on a single driver, proposing demographic, regulatory, and industry-concentration explanations (Decker et al., 2014; Decker et al., 2016; Haltiwanger, Jarmin, and Miranda, 2013; Akcigit and Ates,

2021; Hopenhayn, 1992). None of these explanations has been linked to a technology’s own diffusion schedule in a design that rules out the local and sectoral confounds most likely to contaminate such a claim. We supply a technology-specific mechanism whose timing corresponds to AI’s diffusion: the decline in the incorporated ownership share emerges in 2019, four years after the deep-learning stack became commercial infrastructure, and only in the sectors staffed by AI-exposed occupations.

Third, it speaks to the literature on general-purpose technologies and their historical diffusion through new-firm entry (Bresnahan and Trajtenberg, 1995; Klepper, 2007; Klepper and Sleeper, 2005; Gompers, Lerner, and Scharfstein, 2005). This literature has shown that past general-purpose and high-technology industries, automobiles and lasers among them, were built by spinoff entrants and by incumbent firms that trained future founders. It has documented this entrant-driven diffusion pattern for past technologies but has not tested whether it holds for a technology still in the middle of its diffusion. Babina et al. (2024) suggest AI may depart from this pattern, showing that AI investment today concentrates inside the largest incumbents, and that it is those incumbents, not entrants, who turn the technology into new products and new markets, raising industry concentration rather than seeding new firms. We test whether the same entrant-driven pattern describes AI, using the stock of incorporated business owners as our measure, and we do not find it: incorporated ownership contracts, rather than expands, in the sectors whose occupations AI exposes most, consistent with the task content of exposed occupations narrowing the entrepreneurial channel rather than widening it.

Our sign runs against part of this evidence, and the disagreement is informative rather than incidental. Fossen and Sorgner (2021) separate digitalization that displaces workers from digitalization that complements them, and it is the complementary kind that raises entry into incorporated ownership in their data, because productivity rises and with it the return to founding. That logic requires the technology to complement the founder’s own product. Large language models do not: they raise productivity inside firms while performing the advisory, analytical, and client-facing tasks a small incorporated practice sells, so the same technology that makes an employee more valuable can make an independent practice less viable. Gofman and Jin (2024) document a negative of a different kind, through human capital rather than task content: when AI faculty leave universities, their students go on to found fewer AI startups and raise less funding.

The remainder of the paper proceeds as follows. Section 2 describes the data and the construction of the exposure measure. Section 3 presents the within-commuting-zone research design and its identifying assumptions. Section 4 reports the main result, the unincorporated falsification, and the occupational anatomy behind them. Section 5 collects robustness checks,

the tax-reform horse race, and the timing discussion. Section 6 concludes.

2 Data

2.1 Occupational AI Exposure

Neither firms nor workers are randomly assigned to artificial intelligence. What varies measurably across labor markets is the occupational composition of work, and occupations differ sharply in how much of their task content current AI systems can perform. Our exposure measure aggregates occupation-level AI exposure scores over a baseline occupation mix, following the shift-share logic standard in the local labor markets literature (Autor and Dorn, 2013; Goldsmith-Pinkham, Sorkin, and Swift, 2020).

The unit of treatment is commuting-zone-by-sector level. Commuting zones (CZs) follow the 1990 definitions of Autor and Dorn (2013) and approximate local labor markets; sectors are the 15 Census major industry groups assigned from the ACS industry code. A cell is thus a sector as it exists inside one local labor market, for instance manufacturing in the Detroit commuting zone. The 741 commuting zones and 15 sectors produce 11,112 cells with positive baseline employment, and it is across these cells, within a commuting zone and within a sector, that exposure varies.

Eloundou et al. (2024) build their occupation scores from a task-level rubric applied to the O*NET database of occupational work activities. For each task they ask whether direct access to a large language model, or to software built on one, would cut the time a human needs to complete it by at least half without lowering quality, labeling the task fully exposed, exposed only through complementary software, or unexposed. They collect these labels two ways, from human annotators who know the occupations and from GPT-4 applying the same rubric, and report close agreement between the two, which lets them rate every task of the roughly one thousand O*NET occupations. An occupation’s exposure is then the share of its tasks that AI can materially accelerate; we use their preferred “beta” aggregate, which counts a fully exposed task once and a task exposed only through complementary software at one half, and we map their O*NET-SOC scores to Census occupation codes.

Writing E_o for occupation o ’s resulting Eloundou exposure score, we define

$$AIExposure_{zk} = \sum_o s_{zko,0} E_o, \quad (1)$$

where $s_{zko,0}$ is occupation o ’s share of employment in CZ z and sector k in the 2005–2009 American Community Survey five-year file, the sample the Census Bureau designs for small-

area estimation and thus the appropriate one for an occupation mix measured inside a CZ. The baseline occupation mix is fixed before the estimation window opens in 2010 and predates the deep-learning era entirely, so no outcome observation can inform the shares and the measure cannot respond to post-period changes in local occupational composition. Section 5.1 shows the estimate is unchanged when the baseline is rebuilt on a pre-recession window or on post-recession windows instead. We standardize $AIExposure_{zk}$ to mean zero and unit variance across cells; all estimates are per standard deviation. The Eloundou scores reach 84.9 percent of baseline employment through the O*NET-to-Census occupation crosswalk, and the exposure of a cell is the employment-weighted mean over the workers whose occupations carry a score, while the regression weights count every worker in the cell.

As a robustness measure we build the same aggregate from the AI Occupational Exposure (AIOE) index of Felten, Raj, and Seamans (2021), which links each occupation’s O*NET ability requirements to measured progress of AI systems on those abilities and reaches 98.5 percent of baseline employment. The two measures are constructed by different teams from different primitives, task-level LLM ratings versus ability-level AI benchmarks, and they correlate 0.918 across CZ-sector cells, so the estimates do not turn on the particular way exposure is scored.

Both measures rate what AI *can* do, not what firms actually do, so it is worth asking whether the ranking corresponds to observed adoption. The Census Annual Business Survey asked firms in 2018 how extensively they used artificial intelligence, and 3 percent of employer firms reported any use. Across the 14 sectors the survey and our panel share, that reported use tracks our measure closely: the rank correlation is +0.85 with the Eloundou et al. (2024) aggregate and +0.87 with Felten, Raj, and Seamans (2021), both significant at the 1 percent level (Appendix Table C.1). Information, professional services, and finance sit at the top of both orderings; agriculture and construction at the bottom. The survey validates the ordering of sectors and nothing finer. Its state-by-sector tabulation suppresses half of the cells, and suppression falls on the small, low-exposure sectors, so it cannot speak to the within-sector variation across places that identifies our estimates. We read it as evidence that the task-content rubric picks out where AI is used, not as a first stage.

As an example, Figure 1 maps the measure for a single sector, wholesale trade, across CZs. Holding the sector fixed strips out the between-sector variation that the sector-by-year fixed effects absorb and displays what actually identifies the design: the same industry is substantially more AI-exposed in some local labor markets than in others, because the occupations that staff it differ from place to place. Wholesale trade is the third most exposed of the fifteen sectors, and its exposure ranges from 0.24 to 0.53 across commuting zones.

[Figure 1 about here]

2.2 Measuring Growth-Oriented Entrepreneurship

Entrepreneurship outcomes come from the American Community Survey (ACS) one-year files, 2005–2024, accessed through IPUMS (Ruggles et al., 2024). The ACS class-of-worker variable separates the self-employed into two legal forms: owners of incorporated businesses and unincorporated own-account workers. Levine and Rubinstein (2017) show these two margins contain different people doing different things. The incorporated self-employed hire employees, hold capital, score high on cognitive ability, and earn more than observably similar salaried workers; the unincorporated are closer to subsistence and gig work. We accordingly take the incorporated self-employment rate, incorporated business owners as a share of employed workers aged 16 and over, as our measure of growth-oriented entrepreneurship, and we use the unincorporated rate as a contrast outcome. Pooled over 2005–2024, 3.67 percent of employed American workers are incorporated self-employed and 6.16 percent unincorporated.

We aggregate the person-level microdata to the same commuting zone by sector cells defined above, now by year. We map Public Use Microdata Areas to CZs with crosswalks matched to each PUMA vintage, apportioning PUMAs that straddle CZ boundaries by population weights, and apply ACS person weights throughout. The estimation panel covers 2010–2024, so the pre-2015 comparison years sit inside the post-recession regime and the 2005–2009 files that measure baseline exposure lie entirely outside it; the panel contains 165,495 CZ-sector-year cells with positive employment across 741 CZs. The employment-weighted mean incorporated self-employment rate is 3.71 percent, the denominator for the relative magnitudes we report; the weighted unincorporated mean is 6.05 percent.

Figure 2 shows the raw outcome by sector. The incorporated self-employment rate differs by nearly an order of magnitude across sectors, from roughly 8 percent in construction and agriculture to under 2 percent in education and health, and it is essentially zero in public administration, where incorporation is not an available legal form. The levels are stable through the 2010s and the national aggregate gives no hint of a break, which is the point: whatever AI does to the entry margin happens across commuting zones within a sector, not to sectors as a whole, and the sector-by-year fixed effects of Section 3 discard precisely the variation this figure displays.

[Figure 2 about here]

Table 1 summarizes the estimation panel. Two features of the exposure measure matter for reading the estimates that follow. A standard deviation of exposure, the unit in which every coefficient below is reported, is 0.100 of an occupation’s task share. And essentially all of that variation lies across sectors within commuting zones rather than across them: the

within-commuting-zone standard deviation is 0.099 against 0.100 overall, and 0.032 survives commuting-zone and sector effects together, which is the variation used by the estimator.

[Table 1 about here]

2.3 Nonemployer Statistics

Because the entrepreneurship outcome comes from a household survey, we corroborate it with an independent administrative source. The Census Bureau’s Nonemployer Statistics tabulate, from Internal Revenue Service business tax filings, the universe of U.S. businesses that have no paid employees, classified by legal form of organization and reported by state, sector, and year. The legal-form classification isolates the incorporation margin directly in the tax records: it separates corporations, the C- and S-corporation codes used from 2014 onward together with the single “corporations” code before then, from the sole proprietorships and partnerships that make up the rest. The record has three limits that determine how we use it. Nonemployer businesses have no employees by definition, so it excludes the employer-firm founders that make the incorporated margin a measure of growth entrepreneurship; the legal-form split is published only at the state, not the commuting-zone, level; and the series ends in 2021, three years short of the survey panel, because Nonemployer Statistics are released with a long lag. The register therefore corroborates the decline over a shorter window than the one we estimate. We therefore treat it not as a substitute for the survey but as an independent check on the incorporation margin, and Section 5.3 shows that an IRS-based count of corporation nonemployers falls in AI-exposed sectors alongside the survey measure.

3 Empirical Strategy

To analyze the effects of AI exposure on entrepreneurship, we follow the approach by Johnston and Makridis (2026) by exploiting variation across sectors within commuting zones and absorbing every place-level shock by construction. Let Y_{zkt} be the incorporated self-employment rate (in percentage points) in CZ z , sector k , and year t . The event-study specification is

$$Y_{zkt} = \sum_{\ell \neq -1} \beta_{\ell} (AIExposure_{zk} \times \mathbb{1}[t - T_0 = \ell]) + \delta_{zk} + \mu_{zt} + \eta_{kt} + \varepsilon_{zkt}, \quad (2)$$

where $AIExposure_{zk}$ is the standardized cell-level exposure of equation (1), $T_0 = 2015$ marks the onset of the deep-learning diffusion period, and event time runs from $\ell = -5$ (2010) to $\ell = 9$ (2024) with $\ell = -1$ (2014) omitted. Appendix A sets out the occupational-choice

framework behind this specification: adoption raises both the fixed cost of entry and the salaried return, so the ability threshold for founding rises where exposure is positive and does not move where it is zero. The companion static specification replaces the event-time terms with two phase interactions,

$$Y_{zkt} = \beta_1 (AIExposure_{zk} \times \mathbb{1}[2015 \leq t \leq 2018]) + \beta_2 (AIExposure_{zk} \times \mathbb{1}[t \geq 2019]) + \delta_{zk} + \mu_{zt} + \eta_{kt} + \varepsilon_{zkt}, \quad (3)$$

splitting the post-period into an onset window (2015–2018) and a diffusion window (2019–2024).

The two windows summarize a dynamic path rather than replace it. Equation (2) is the primary specification, and equation (3) pools its coefficients into two numbers that can be reported with a standard error. Both boundaries follow the technology. We date the onset to 2015 because that is when the deep-learning stack became free infrastructure: after the research breakthroughs of 2012–2015, Google open-sourced TensorFlow in November 2015, putting frontier machine-learning tools in general hands. The 2015–2018 window then covers initial adoption and the complementary investment it requires, a period over which firm-level AI investment scaled (Babina et al., 2024), and during which the entry margin should not yet have moved much. The 2019 boundary marks the first year the event-study path turns persistently and economically negative, after that investment ramp-up and before either COVID-19 or any public large-language-model product. We are explicit that the 2019 split is a parsimonious summary of the estimated path and of external diffusion evidence, not a break we pre-specified or that an estimator selects on its own: a structural-break search over the coefficient path is consistent with it but cannot resolve the exact year, locating the break in 2019 with a confidence interval spanning 2016–2020 (Section 5.2). What that search does settle is that the break is not in 2018, the year a tax-driven explanation would require. Alternative 2018 and 2020 boundaries leave the estimate essentially unchanged and are reported in Appendix Table C.5.

We estimate both specifications by OLS, weighting cells by baseline (2005–2009) employment and clustering standard errors at the CZ level, the coarsest level at which exposure components are correlated. The fixed effects do the identification work, and each one closes a specific door. The CZ-sector effects δ_{zk} absorb permanent specialization: Silicon Valley’s professional services and Houston’s mining can differ permanently in both exposure and entrepreneurship without affecting the estimates. The sector-year effects η_{kt} absorb national sectoral shocks, including anything AI does to a sector everywhere at once; identification uses only the differential occupational exposure of the same sector across local labor markets. The

CZ-year effects μ_{zt} are the decisive margin. They absorb every shock to a commuting zone in a year, whatever its source: the local college share and its trend, agglomeration economies, housing costs, local credit conditions, pandemic-era shocks, and any boom or bust specific to educated superstar cities. The concern that AI exposure merely proxies for educated, urban places, and that the technology travels with the college share because it complements skilled cognitive work, is addressed here by construction, as places are compared only with themselves within a year. Nothing measured at the level of the commuting zone, in any year, survives to confound the estimate.

The identifying assumption that remains is parallel trends within place: absent AI, more- and less-exposed sectors inside the same commuting zone would have trended alike, conditional on national sector trends. The assumption has a five-year testable analogue, and it passes: the pre-2015 coefficients of equation (2) are jointly indistinguishable from zero in every specification we report ($F = 0.39$, $p = 0.81$ in the main design; Section 5), with the usual caution that pre-trend tests have finite power (Roth, 2022).

The remaining threat would be a shock arriving after 2015 that hits high-exposure sectors differentially in some commuting zones but not others, tracks the occupational exposure gradient rather than the sector, and does so on AI’s timetable. The one concrete candidate we can name is the 2017 Tax Cuts and Jobs Act, whose Section 199A pass-through deduction singled out many of the same professional occupations; Section 5.4 addresses it directly and finds the estimate unmoved and its timing wrong for a tax story. Beyond that we know of no candidate other than AI itself. Because exposure is a continuous treatment, the estimates average responses along the exposure gradient and carry a causal interpretation under the parallel-trends condition of Callaway, Goodman-Bacon, and Sant’Anna (2024); the design’s shift-share treatment rests on the exogeneity of baseline occupation shares in the sense of Goldsmith-Pinkham, Sorkin, and Swift (2020), fixed over 2005–2009, before the estimation window opens.

4 Results

We discuss the results in three steps. We first present the event-study and two-phase estimates for the incorporated self-employment rate, together with the unincorporated contrast showing that the decline is concentrated on the incorporated margin rather than self-employment broadly (Section 4.1). We then turn to the occupations of the entrepreneurs themselves, which show why the growth-oriented margin was exposed to AI’s task content and tie the sector-level decline to the task content AI performs (Section 4.2). We close by decomposing exposure into its automating and augmentative components, which carry the two phases of

the decline separately and connect it to independent evidence on employer-side outcomes (Section 4.3).

4.1 The Decline in Incorporated Self-Employment

Figure 3 reports the paper’s central result: event-study estimates of equation (2) for the incorporated self-employment rate. In the five years preceding 2015, sectors with differing AI exposure within the same commuting zone follow similar trends; the pre-period coefficients range from +0.05 to +0.13 percentage points and are individually and jointly indistinguishable from zero. Following 2015, the coefficients diverge in two phases. Between 2015 and 2018 the estimates are mildly negative, ranging from -0.02 to -0.11 percentage points, none individually significant, although the pooled onset-window estimate is significant at the 5 percent level (Table 2). Beginning in 2019 the coefficients become larger in magnitude and statistically significant: -0.22 percentage points per standard deviation of exposure (significant at the 10 percent level), followed by -0.43 in 2020, -0.42 in 2021, -0.27 in 2022, -0.35 in 2023, and -0.30 in 2024. The estimates are statistically significant at the 10 percent level or higher from 2020 onward.

[Figure 3 about here]

Table 2 reports the two-phase estimates of equation (3). In the main specification (column 1), a one-standard-deviation increase in AI exposure lowers the incorporated self-employment rate by 0.144 percentage points in 2015–2018 (standard error 0.068), significant at the 5 percent level, and by 0.401 percentage points in 2019–2024 (standard error 0.088), significant at the 1 percent level. Put differently, in 2019–2024, a commuting-zone-by-sector cell with AI exposure one standard deviation higher experienced a 0.401-percentage-point larger decline in incorporated self-employment relative to 2010–2014, after accounting for commuting-zone-by-year and sector-by-year shocks. Against the employment-weighted mean rate of 3.71 percent, this represents a 10.8 percent decline, which is approximately one fewer incorporated business owner per 250 employed workers. This estimate is robust to alternative exposure measures and sample restrictions: the Felten, Raj, and Seamans (2021) measure yields -0.443 percentage points in 2019–2024 (column 2), excluding the agriculture sector yields -0.412 (column 3), and restricting to cells with baseline employment of at least 5,000 workers yields -0.454 (column 4). Joint tests of pre-trends fail to reject in every column.

[Table 2 about here]

Panel B examines whether this decline is specific to the growth-oriented margin of self-employment. For incorporated self-employment, the estimated effect increases in magnitude as

the technology diffuses: from -0.144 percentage points in the onset window to -0.401 in the diffusion window under Eloundou et al. (2024), and from -0.104 to -0.443 under Felten, Raj, and Seamans (2021). Tests of equality between the two windows reject under both measures, with p-values of 0.002 and below 0.001, respectively. The unincorporated margin exhibits no comparable pattern. Under our primary measure, the estimated coefficients in both windows are statistically indistinguishable from zero, and pre-trends do not reject ($F = 1.54$, $p = 0.19$). The Felten measure does detect a decline in unincorporated self-employment, but the estimated coefficient is nearly identical across windows, -0.203 in the onset period versus -0.212 in the diffusion period, and a test of equality between them returns a p-value of 0.97. This pattern, an immediate shift in 2015 that does not deepen thereafter, is inconsistent with a mechanism operating through a technology whose adoption continued to scale through 2018.

Because incorporated and unincorporated self-employment have different mean rates (3.71 and 6.05 percent, respectively), we express each estimate relative to its own mean. In 2019–2024, the incorporated rate declines by 10.8 percent under the Eloundou et al. (2024) measure and 11.9 percent under the Felten, Raj, and Seamans (2021) measure. The corresponding declines in the unincorporated rate are 1.3 and 3.5 percent. The proportional decline is therefore approximately eight times larger for incorporated self-employment under Eloundou et al. (2024) and three times larger under Felten, Raj, and Seamans (2021). This asymmetry weighs against explanations that should affect both legal forms similarly. It is confined to the diffusion window: during 2015–2018, the two margins are not clearly separated.

Figure 4 makes this divergence visible. The incorporated coefficients turn sharply negative during the diffusion period and remain negative through 2024, including the LLM period, whereas the unincorporated coefficients remain statistically indistinguishable from zero throughout. The decline is therefore concentrated among incorporated owners, the group that Levine and Rubinstein (2017) show is more likely to hire employees and invest capital. AI-exposed sectors experience a decline in growth-oriented business ownership rather than a broad decline in self-employment.

[Figure 4 about here]

This magnitude cannot be attributed to place-specific shocks. Because the commuting-zone-by-year fixed effects absorb all such shocks by construction, the estimated decline cannot reflect a general souring of entrepreneurial activity in highly educated cities, housing cost pressures specific to technology hubs, or differential exposure to the pandemic across localities. The effect instead operates through the exposed sectors themselves, within a given commuting zone and year alike.

4.2 Who Are the Exposed Entrepreneurs?

The occupational composition of incorporated entrepreneurs helps explain why the shock is concentrated in growth-oriented firms. Figure 5 plots each occupation’s incorporated self-employment rate against its Eloundou et al. (2024) exposure score across the 408 occupations of the pooled 2010–2019 ACS (the figure displays the 273 occupations with pooled employment above 500,000 for legibility; the correlations reported below use all 408). The relationship is positive (Table C.4): the employment-weighted correlation between an occupation’s incorporated self-employment rate and its AI exposure is +0.20 with the Eloundou score and +0.23 with the Felten score, while the unincorporated rate correlates negatively with both (−0.17 and −0.15). The occupations that produce incorporated entrepreneurs are managerial and professional: chief executives (a 24 percent incorporated self-employment rate, at the 79th percentile of exposure), real-estate brokers (20 percent, 71st), and lawyers (14 percent, 70th), followed by management analysts, construction and food-service managers, and sales supervisors. In aggregate, the top tercile of occupations by exposure accounts for 56.0 percent of all incorporated self-employment while comprising 37.7 percent of employment; the bottom tercile accounts for 17.9 percent of incorporated self-employment on 24.8 percent of employment. This composition is measured over 2010–2019, prior to the steepest years of the estimated decline and prior to the pandemic, and therefore describes the pre-existing occupational structure of incorporated entrepreneurship rather than an outcome of the treatment period.

[Figure 5 about here]

This composition is descriptive. The design does not observe AI adoption, separate business entry from exit, or follow former and prospective owners to their subsequent jobs, so it cannot establish which occupational characteristic carries the decline. Establishing that requires linked worker-owner-employer records combined with direct measures of firm-level AI adoption, so that future research can trace exposed workers into ownership entry, ownership exit, and salaried employment.

Two additional analyses are consistent with this occupational pattern. Appendix Table C.10 shows that the estimated decline is broad-based across sectors rather than concentrated in a single industry: estimated separately for each of the fifteen sectors with all fixed effects retained, the 2019–2024 coefficient is negative in all sectors except public administration and significant at the 10 percent level or better in eight, spanning construction, wholesale trade, professional and business services, education and health, information, utilities, and manufacturing. In agriculture, where self-employment largely reflects farm proprietorship rather than firm formation, the estimated coefficient is economically small and

statistically insignificant.² Appendix Table C.11 shows that across age groups, the estimated decline is concentrated among established workers rather than young workers: the effect is present at every age from 30 upward, largest among the 45–59 age group, and statistically indistinguishable from zero only for workers under 30, an age group in which incorporation is uncommon to begin with. This pattern is consistent with a decline concentrated among established professionals rather than first-time founders early in their careers, distributed across the sectors that employ those professionals.

4.3 Automating and Augmentative AI Exposure

Eloundou et al. (2024) separate tasks a large language model completes on its own from tasks where the same speedup requires complementary software or human collaboration. Following Massenkoff and McCrory (2026) and Johnston and Makridis (2026) we call the first automating exposure and the second augmentative exposure. We rebuild the cell-level measure separately for each using the same baseline occupation shares and the same score-matched denominator, and we enter both in the main specification jointly. Summing the two components recovers the published exposure measure exactly, up to floating-point rounding of 2.4×10^{-15} .

The two components carry the two phases (Table C.3). The incorporated self-employment rate falls 0.279 percentage points per standard deviation of automating exposure during 2015–2018, and the coefficient does not deepen thereafter, a diffusion-minus-onset difference of +0.115 ($p = 0.16$), so automating exposure accounts for the onset window. Augmentative exposure runs the other way. It is indistinguishable from zero during the onset window and falls 0.283 percentage points per standard deviation during 2019–2024, a deepening of 0.357 points ($p < 0.001$). Pre-2015 leads are jointly indistinguishable from zero for both components, with p-values of 0.77 and 0.39 (Figure C.1).

As in Section 5.2, a response present at the onset that does not deepen is inconsistent with a technology whose adoption continued to scale. On that criterion the diffusion-window decline is carried by augmentative exposure rather than by automating exposure. Johnston and Makridis (2026) find, on the employer side, that augmentative exposure raises sector employment by 3.6 percent and the count of establishments by 3.3 percent by 2024, while automating exposure moves neither. The component of AI exposure that expands employer firms is the component that contracts incorporated business ownership. Appendix B reports what the same exposure measure does to employee wages in our own cells, and why a cell

²These sector-specific coefficients come from a triple interaction on the full panel that keeps the commuting-zone-by-year fixed effects, so each is identified from cross-commuting-zone variation within the sector; the estimand differs from the pooled coefficient and the magnitudes are not directly comparable to it.

mean and their within-occupation estimate need not agree in sign.

We read this decomposition as descriptive rather than as an identified channel. Three features of the exhibits urge caution. The two components are built from a single occupational score and correlate 0.58 across commuting-zone-by-sector cells, so they are not independent treatments at this level of aggregation. The augmentative diffusion-window estimate rests heavily on one year: the coefficient is -0.51 in 2020 and between -0.10 and -0.15 in each of 2021 through 2024. And the unincorporated margin, which the aggregate exposure measure leaves unchanged, does respond to the augmentative component, by -0.275 percentage points. Against the two means the proportional declines are 7.6 percent for incorporated ownership and 4.5 percent for unincorporated, so the asymmetry of Section 4.1 survives the decomposition but is narrower within it.

5 Robustness

This section addresses four sources of concern in turn. Section 5.1 shows that the main estimate is robust to alternative measurement and sample choices. Section 5.2 addresses the four-year lag between the 2015 onset and the 2019 response and shows that this lag is consistent with the predicted timing of technology diffusion. Section 5.3 corroborates the survey-based decline using administrative tax records. Section 5.4 tests the one rival explanation that shares our identifying variation directly against the AI exposure measure: the Section 199A pass-through deduction introduced by the 2017 tax reform.

5.1 Robustness of the Main Estimate

Beyond the alternative exposure measures, sample restrictions, and unincorporated-margin falsification test already presented alongside the main estimate in Section 4.1, five additional features of the design merit attention.

First, the result does not depend on the fifteen-sector aggregation. Replacing the major sectors with the 223 harmonized detailed industries of the IPUMS IND1990 classification leaves pre-trends flat ($F = 0.08$ to 0.57 across employment floors) and puts the 2019 coefficient between -0.25 and -0.54 percentage points, distinguishable from zero across floors (Figure C.2); precision falls as cells thin at the highest employment floor, but the estimated decline is if anything larger at finer granularity, indicating that the sector definition does not drive the result.

Second, the estimate does not depend on the window used to construct exposure. The 2005–2009 baseline lies entirely outside the estimation panel but spans the Great Recession,

raising the concern that a recession-distorted occupation mix drives the measure. Table 3 rebuilds exposure on a pre-recession window (2005–2007) and two post-recession windows (2010–2012, 2010–2013); every column is negative and significant at the 1 percent level, running from -0.324 to -0.461 against -0.401 in the main design, with flat pre-trends throughout, and the four windows’ exposure indices correlate between 0.986 and 0.996 employment-weighted.

The recession itself does not drive the spread: holding window length fixed at three years and varying only whether the recession falls inside it gives estimates of -0.324 , -0.313 , and -0.336 , a spread of 0.02 percentage points against a standard error of 0.08. The spread across columns instead reflects sample size. Because the commuting-zone-by-sector fixed effects absorb the level of exposure, only the variation net of commuting-zone and sector effects identifies the estimate, and shorter windows carry more sampling error in exactly that residual; classical measurement-error attenuation accounts for the three-year windows’ smaller coefficients, and single-year shares attenuate further, to between -0.20 and -0.30 . We accordingly build exposure on the longest pre-sample window available, the 2005–2009 five-year file, which is also the sample the Census Bureau designs for small-area estimation. Post-recession windows carry the opposite caution: because they overlap the panel’s first years, their occupation shares and the outcome share the same sampled workers, which mechanically inflates early pre-period coefficients and, through them, the diffusion estimate.³

Third, the decline does not reflect sectoral reallocation within the commuting zone. Because μ_{zt} absorbs each commuting zone’s total employment in every year, a genuine decline in growth entrepreneurship and a reallocation of growth entrepreneurship across sectors within a place are observationally equivalent under the main specification. Dropping μ_{zt} distinguishes between these possibilities, since pure reallocation would cancel out in the outcome itself, while all other features of the specification, cells, weights, clustering, and sector-by-year effects, are held fixed. The estimate remains stable under this alternative specification: the 2019–2024 coefficient is -0.453 (standard error 0.071) against -0.401 in the main specification, pre-trends remain flat ($F = 0.53$, $p = 0.71$), and the same pattern holds under Felten, Raj, and Seamans (2021) (-0.399) and excluding agriculture (-0.453 ; Table 4). Absorbing every place-year shock moves the point estimate by only 0.05 percentage points, well within its standard error. The commuting-zone-by-year fixed effects therefore do not drive the magnitude of the estimate; rather, they rule out place-level confounds and support a causal interpretation of the result.

³The baseline must be pre-sample, not merely pre-treatment. Building shares from 2014, the last pre-onset year, makes the exposure and reference-period outcome share the same sampled workers, and the pre-2015 coefficients then reject parallel trends ($F = 5.5$, $p < 0.001$), against $F = 0.39$ on the pre-sample baseline.

Three additional checks, documented in Appendix C, are summarized here. No small set of occupations is responsible for the exposure measure: rebuilding it without the top Rotemberg-weighted contributors leaves the 2019–2024 estimate between -0.393 and -0.420 , and the highest-weight occupations are broadly distributed clerical and sales occupations rather than elite professions (Appendix Tables C.7–C.8). The baseline shares are balanced on pre-trends in the sense the design requires: they predict raw cross-sectional trends ($F = 12.63$) but lose all predictive power once commuting-zone and sector effects are absorbed ($F = 0.73$, $p = 0.70$; Appendix Table C.9, Panel A). The decline also does not fade over time: the 2019–2024 event-study coefficients are jointly indistinguishable from one another ($\chi^2_5 = 3.23$, $p = 0.66$) and exhibit no linear trend (Appendix Table C.9, Panel B).

Fourth, we address inference. A shift-share regressor induces correlated residuals across cells that share an occupation mix, an issue that clustering by commuting zone does not address. We therefore recompute the standard error using the exposure-robust variance estimator of Adão, Kolesár, and Morales (2019), applied to the panel collapsed to the long difference that the two-phase specification identifies. The collapsed point estimate is -0.405 , with an exposure-robust standard error of 0.068 , compared with 0.088 when clustering by commuting zone (Appendix Table C.2). Correcting for the shares tightens the confidence interval rather than widening it, so the inference reported throughout the paper is the more conservative of the two.

[Table 3 about here]

[Table 4 about here]

5.2 Timing and the Diffusion Lag

The estimated effect emerges in 2019, four years after the 2015 onset that anchors the event studies. Section 3 explains the timing: the deep-learning stack became widely accessible infrastructure in 2015, and the lag to a measurable entry response matches the predicted pace of technology diffusion, since firm-level AI investment scaled only over 2016–2018 (Babina et al., 2024) and the entry margin should respond once founders face AI-capable incumbents, not when the underlying research appears. We test this timing directly rather than assume it: a structural-break test on the event-study coefficient path, a sup-Wald search over candidate onset years with a bootstrap confidence interval, rejects the no-break null decisively (sup-Wald statistic of 23.5 against a 5 percent critical value of 8.85), locates the break in 2019 (95 percent CI: 2016–2020), and rules out 2018, the year a tax-driven reorganization would require, with only 4 percent of bootstrap draws selecting it. The headline estimate is insensitive to whether

the diffusion window is dated from 2018, 2019, or 2020, giving -0.365 , -0.401 , and -0.424 , respectively (Appendix Table C.5).

The fact that our exposure measure is derived from large language models, which reached the public only after the estimated decline begins, does not undermine the timing of the proposed mechanism. The measure ranks occupations by the cognitive-task content that AI can automate; it is not tied to the release of any single product, and the ranking does not depend on the existence of language models. The Felten, Raj, and Seamans (2021) exposure measure, built from the overlap between occupational abilities and the broader frontier of AI progress through the 2010s rather than from language-model ratings, correlates 0.918 with our measure across cells and delivers a nearly identical estimate (-0.443 ; column 2 of Table 2). The set of exposed occupations is essentially the same whether exposure is measured using language-model capabilities or the earlier, broader machine-learning frontier, and it is that broader frontier, the mid-2010s deep-learning inflection and the scaling of firm-level AI investment over 2016–2018 (Babina et al., 2024), whose timing the estimated onset tracks, roughly half a decade before commercial language models became available. We accordingly date the diffusion window from 2019 rather than from ChatGPT’s release in late 2022: the 2019 onset precedes any large-language-model product by three years, and with the panel ending in 2024, a 2022 onset would leave only a single post-treatment year, insufficient for identification.

The estimated decline does not appear to be driven by the COVID-19 pandemic. The decline is already evident in 2019 (-0.22 percentage points, significant at the 10 percent level), which predates the onset of COVID-19, and the pooled 2019–2024 estimate is significant at the 1 percent level. The estimates for 2021–2024 (-0.42 , -0.27 , -0.35 , -0.30) persist through years in which U.S. business formation was booming nationally, and the design absorbs pandemic-related shocks by construction: commuting-zone-by-year fixed effects remove local pandemic severity and policy responses, and sector-by-year fixed effects remove national sectoral shocks, including the pandemic’s uneven incidence across sectors.

One measurement issue deserves note. The ACS measures entrepreneurship from survey reports, not business registers, and a register of establishment *births* (the Business Dynamics Statistics) fails parallel pre-trends here, since high-exposure sectors were already on differential birth trajectories before 2015, unlike the incorporation *stock*. Section 5.3 takes up the register evidence on this stock directly; throughout, our outcome should be read as a stock share rather than a flow of firm births.

5.3 Survey and Administrative Records

Because the estimates come from a household survey, the distinction between incorporated and unincorporated could in principle reflect self-report error rather than real business form. Table 5 addresses this in two steps. First, the effect is not specific to commuting-zone geography: aggregating the same design to the state level leaves the 2019-onward estimate intact (-0.405 percentage points, significant at the 5 percent level; column 2) on flat pre-trends ($F = 0.66$). Second, we replace the survey with the Nonemployer Statistics tax records introduced in Section 2.3. We measure the incorporated margin there as corporation nonemployers per thousand workers, the same per-worker normalization as the survey rate, with employment drawn from the same state-sector-year cells, and it falls sharply in AI-exposed sectors after 2015 (-2.78 per thousand workers in 2019 onward, significant at the 1 percent level; column 3) on flat pre-trends ($F = 1.24$). Figure 6 traces the year-by-year path: the pre-2015 coefficients are close to zero, and the decline begins after the 2015 onset and deepens through 2021.

This per-worker normalization is essential to a valid comparison. The raw corporation count fails the parallel-trends test because high-exposure sectors were adding workers, and businesses of every legal form, more rapidly before 2015, but dividing by employment removes exactly that common growth: $\log(\text{corp}) - \log(\text{emp})$ is flat before 2015, whereas $\log(\text{corp})$ alone is not. Because contemporaneous employment is itself exposed to AI, column 4 divides the same corporation counts by *baseline* employment, the fixed 2005–2009 state-sector measure used to weight every other specification in the paper, a denominator determined before the estimation window opens and therefore exogenous to the treatment; the estimated decline is if anything larger (-3.35 per thousand baseline workers in 2019 onward, significant at the 5 percent level), and pre-trends still do not reject ($F = 1.61$), although fixing the denominator makes the pre-period drift more visible, since it no longer nets out the sectors’ own employment growth. The IRS-based count and the household survey thus decline together, and significantly so, in the one geography where both sources observe incorporation, an outcome that a coding artifact in either source would be unlikely to produce.

As Section 2.3 notes, the nonemployer records exclude employer-firm founders, so the register corroborates the sign, magnitude, and significance of the decline in incorporated business activity using independent administrative data. However, the falsification test contrasting incorporated with unincorporated self-employment (Section 5.1) can only be conducted using the survey, since the survey alone observes owners of employer firms.

[Table 5 about here]

[Figure 6 about here]

5.4 Tax Reform and the Section 199A Exclusion

The most plausible alternative explanation for our results is a tax-policy channel. The 2017 Tax Cuts and Jobs Act, effective in tax year 2018, granted pass-through businesses a 20 percent deduction on qualified income under Section 199A, but above certain income thresholds it denied that deduction to “specified service trades or businesses”: health, law, accounting, actuarial science, performing arts, consulting, athletics, financial services, brokerage, and investing. This list overlaps substantially with our exposed occupations: lawyers, consultants, financial advisers, and physicians are all highly AI-exposed and were all denied the deduction. If the reform reduced incorporated pass-through activity in service-heavy cells relative to other cells within the same commuting zone, this effect would load onto the same occupational variation used in our design, and neither set of fixed effects would absorb it. This is the only confounder we are aware of that shares our identifying variation.

We therefore construct a specified-service exposure measure, $SSTBShare_{zk}$, following exactly the construction of equation (1): the baseline 2005–2009 employment share, in each commuting-zone-by-sector cell, of the thirty-nine occupations we map to Section 199A service fields.⁴ The two measures are only moderately correlated, 0.52 with the Eloundou measure and 0.70 with the Felten measure, because the statute exempts engineering and architecture despite both being highly AI-exposed, and because many exposed clerical and sales occupations are not service occupations at all. This leaves sufficient independent variation to distinguish between the two measures.

Table 6 reports the results. Adding $SSTBShare_{zk}$ interacted with the phase indicators leaves the AI coefficient at -0.407 , compared with -0.401 without it, while the service share itself is estimated at 0.017 (standard error 0.076). Entered alone, in column 3, the service share still does not predict the incorporation decline. Splitting cells at the employment-weighted within-sector median of service share and estimating the AI effect separately on each side gives -0.426 where the tax exclusion binds and -0.427 where it does not. The decline is equally large in low-service cells, a pattern that no Section 199A mechanism can produce.

The timing evidence points to the same conclusion. The deduction takes effect in tax year 2018, and the ACS records current class of worker, so a tax-driven reorganization should be evident in the 2018 data. AI diffusion, given that firm-level investment scaled over 2016–2018,

⁴The mapping follows the statutory field list of Internal Revenue Code Section 199A(d)(2) and the Treasury regulations: health practitioners, lawyers and legal support, accountants and tax preparers, actuaries, financial advisers and analysts, management consultants, performing artists, and athletes and coaches. Engineering, architecture, banking, insurance, and real-estate brokerage are excluded by statute; the full occupation-level crosswalk is in the replication package.

predicts a later break. The 2018 coefficient is -0.081 and statistically indistinguishable from zero; the estimated decline instead begins in 2019, consistent with the structural-break estimate, which places only 4 percent probability on 2018 (Section 5.2). Appendix Figure C.3 plots both coefficient paths by year: the AI-exposure path is flat before 2015 and declines after 2019, while the service-share path remains close to zero throughout and drifts slightly positive after 2018.

An additional check considers the opposite implication. Section 199A preserved the deduction for below-threshold owners, who make up most of the unincorporated margin, so a reform that reshaped legal-form choice should *raise* unincorporated self-employment in service-heavy cells; column 4 reports a coefficient of -0.006 . Two further features of the reform work against our estimate rather than for it: the same legislation cut the C-corporation tax rate from 35 to 21 percent, which makes incorporation more attractive, and the service exclusion binds only above the income thresholds, exempting most own-account owners. Rebuilding both shift-share measures on the pre-recession 2005–2007 baseline leaves the AI coefficient at -0.323 and the service-share coefficient at -0.001 (Appendix Table C.6). We interpret the incorporation decline as attributable to AI exposure rather than to the 2017 tax reform.

[Table 6 about here]

6 Conclusion

The entrepreneurial channel through which past general-purpose technologies diffused appears to be closing rather than opening under artificial intelligence. Within commuting zones, sectors with higher AI exposure have an incorporated self-employment rate 0.401 percentage points lower per standard deviation of exposure in 2019–2024, a 10.8 percent decline, while unincorporated self-employment shows no comparable change. The decline is concentrated in the exposed sectors themselves, not in the places that host them, and the founders it removes are disproportionately drawn from the managerial and professional occupations whose task content large language models increasingly perform. Babina et al. (2024) document the corresponding firm-level pattern, AI investment concentrating in large incumbents; our results are the analogous pattern in labor markets.

These results speak to two policy and research debates. If AI’s economics favor incumbents at the entry margin, policies that subsidize invention alone will not reopen the entrepreneurial channel. And they offer the declining-dynamism literature (Decker et al., 2014; Decker et al., 2016) a technology-specific candidate driver, since the decline begins in AI-exposed sectors on a timing broadly consistent with the technology’s diffusion after 2015 (Section 5.2).

Three limitations apply. Occupational exposure measures what AI can do, not firm-level adoption. Our outcome is a stock share of self-employed individuals rather than firm entry, exit, or growth directly; administrative tax records corroborate the stock decline (Section 5.3), but we cannot yet trace it to firm births. And our sample extends only into 2024, two years into the diffusion of generative AI.

If the mechanism continues, the same four-year lag would put a comparable response around 2026, following generative AI’s commercial-scale diffusion after ChatGPT’s 2022 release, though this remains untested in our data. Whether the decline reflects slower entry, faster exit, or both awaits linked employer-employee data. Appendix A rules out the incumbent-wage channel but cannot separate a rising entry-scale requirement from an erosion in the returns to firm ownership; distinguishing them requires establishment-level microdata. Whether AI eventually reproduces the broad diffusion of past general-purpose technologies, or continues to narrow the entrepreneurial margin, remains open.

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Table 1: Summary statistics

	Mean	Emp.-wtd mean	SD	P10	P50	P90
<i>Panel A: Outcomes</i>						
Incorporated self-employment rate (%)	4.12	3.71	5.32	0.00	2.43	10.44
Unincorporated self-employment rate (%)	8.35	6.05	10.86	0.00	3.97	22.91
<i>Panel B: AI exposure, raw scale</i>						
Eloundou (share of tasks)	0.32	0.35	0.10	0.19	0.32	0.46
Felten (AIOE index)	-0.11	0.07	0.49	-0.69	-0.15	0.55
<i>Panel C: Cell size</i>						
Cell employment, 2005–2009 (mean annual)	12,782		47,245	178	2,087	25,044
<i>Memo: standard deviation of Eloundou exposure</i>						
Overall (the unit of the reported coefficients)	0.100					
Within commuting zone	0.099					
Net of CZ and sector effects	0.032 (32% of overall)					

Notes: Cell-level distributions across the 165,495 commuting-zone $\times$ sector $\times$ year cells of the 2010–2024 estimation panel (741 commuting zones, 15 Census sectors). The incorporated and unincorporated self-employment rates are cell shares of employed workers aged 16 and over (ACS CLASSWKRD codes 14 and 13). The employment-weighted means use baseline (2005–2009) mean annual cell employment, the fixed weight used in estimation; these are the denominators for the relative magnitudes reported in the text, and they fall below the unweighted means because small cells are numerous and have high self-employment rates. Standard deviations and percentiles are unweighted. Panel B reports both exposure measures on their raw scale rather than standardized: the Eloundou et al. (2024) aggregate is a share of an occupation’s tasks, averaged over the cell’s 2005–2009 occupation mix by equation (1), and the Felten, Raj, and Seamans (2021) AIOE index is a standardized ability-based score with no natural units. An employment-weighted mean of cell employment is not reported because it is not an interpretable quantity. The memo panel gives the dispersion of Eloundou et al. (2024) exposure across the 11,112 commuting-zone $\times$ sector cells: the overall standard deviation of 0.100 is the unit in which every coefficient in the paper is expressed, the within-commuting-zone figure shows that essentially all of the variation is across sectors inside places rather than across places, and the figure net of commuting-zone and sector effects is the variation used by the estimator once the fixed effects absorb the level.

Table 2: AI exposure and self-employment incorporation

Exposure measure Sample	Eloundou All (1)	Felten All (2)	Eloundou Excl. agric. (3)	Eloundou Emp $\geq$ 5k (4)
<i>Panel A: Incorporated self-employment rate</i>				
AI exposure $\times$ 2015–2018	-0.144** (0.068)	-0.104 (0.066)	-0.169*** (0.066)	-0.133 (0.082)
AI exposure $\times$ 2019–2024	-0.401*** (0.088)	-0.443*** (0.083)	-0.412*** (0.088)	-0.454*** (0.107)
<i>Panel B: Unincorporated self-employment rate</i>				
AI exposure $\times$ 2015–2018	-0.107 (0.095)	-0.203** (0.085)	-0.078 (0.093)	-0.089 (0.112)
AI exposure $\times$ 2019–2024	-0.078 (0.104)	-0.212** (0.099)	0.011 (0.105)	-0.016 (0.126)
Observations	165,495	165,495	154,441	51,498
CZ $\times$ sector FE	Yes	Yes	Yes	Yes
CZ $\times$ year FE	Yes	Yes	Yes	Yes
Sector $\times$ year FE	Yes	Yes	Yes	Yes

Notes: Unit of observation: commuting zone $\times$ sector $\times$ year (741 commuting zones, 15 Census sectors, 2010–2024). Coefficients on $AIExposure_{zk} \times \mathbb{1}[2015 \leq t \leq 2018]$ and $AIExposure_{zk} \times \mathbb{1}[t \geq 2019]$ from equation (3), estimated by OLS with commuting-zone-by-sector, commuting-zone-by-year, and sector-by-year fixed effects, weighted by baseline (2005–2009) mean annual cell employment. Panel A reports the incorporated self-employment rate in percentage points and Panel B the unincorporated rate; the two panels share the same four column specifications and the same estimating sample. Exposure is the standardized Eloundou et al. (2024) measure of equation (1) except in column 2, which uses the Felten, Raj, and Seamans (2021) AIOE aggregate. Column 3 excludes the agriculture sector; column 4 restricts to cells with at least 5,000 baseline workers in an average year. Standard errors clustered by commuting zone in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Robustness to the AI-exposure baseline window

	(1)	(2)	(3)	(4)
AI exposure $\times$ 2015–2018	-0.144** (0.068)	-0.083 (0.067)	-0.200*** (0.066)	-0.195*** (0.066)
AI exposure $\times$ 2019–2024	-0.401*** (0.088)	-0.324*** (0.082)	-0.456*** (0.085)	-0.461*** (0.086)
Observations	165,495	165,425	165,518	165,529
Exposure baseline window	2005–2009	2005–2007	2010–2012	2010–2013
Character	Pre-sample	Pre-recession	Overlaps panel	Overlaps panel
CZ $\times$ sector FE	Yes	Yes	Yes	Yes
CZ $\times$ year FE	Yes	Yes	Yes	Yes
Sector $\times$ year FE	Yes	Yes	Yes	Yes

Notes: Two-phase difference-in-differences of the incorporated self-employment rate on standardized occupational AI exposure interacted with post-2015 indicators, reference years 2010–2014, at the commuting-zone $\times$ sector $\times$ year level (2010–2024). Columns differ only in the window whose workforce defines the exposure shares and the fixed regression weights. Column 1 is the 2005–2009 baseline used throughout the paper; columns 2–4 rebuild the measure from the ACS one-year files on a pre-recession window and two post-recession windows. All columns cluster by commuting zone and saturate with commuting-zone-by-sector, commuting-zone-by-year, and sector-by-year fixed effects. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: The commuting-zone-by-year margin

	Incorporated self-employment rate			
	(1)	(2)	(3)	(4)
AI exposure $\times$ 2015–2018	-0.1439** (0.0680)	-0.0704 (0.0529)	-0.0341 (0.0448)	-0.0718 (0.0515)
AI exposure $\times$ 2019–2024	-0.4014*** (0.0876)	-0.4526*** (0.0712)	-0.3990*** (0.0605)	-0.4531*** (0.0710)
Observations	165,495	165,495	165,495	154,441
R ²	0.76377	0.74521	0.74524	0.75935
Within R ²	0.00050	0.00099	0.00112	0.00112
CZ $\times$ sector fixed effects	✓	✓	✓	✓
CZ $\times$ year fixed effects	✓			
Sector $\times$ year fixed effects	✓	✓	✓	✓

Notes: Two-phase difference-in-differences of equation (3) at the commuting-zone $\times$ sector $\times$ year level (741 commuting zones, 15 Census sectors, 2010–2024). Column 1 is the main specification; columns 2–4 remove the commuting-zone-by-year fixed effects μ_{zt} and change nothing else. Column 2 uses Eloundou et al. (2024), column 3 Felten, Raj, and Seamans (2021), column 4 excludes agriculture. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Survey and administrative records: the incorporation margin at coarser geography

	(1)	(2)	(3)	(4)
AI exposure $\times$ 2015–2018	-0.144** (0.068)	-0.161 (0.125)	-1.232** (0.502)	-1.696** (0.705)
AI exposure $\times$ 2019 onward	-0.401*** (0.088)	-0.414** (0.190)	-2.776*** (0.824)	-3.346** (1.258)
Observations	165,495	11,456	8,467	8,467
Data source	ACS (survey)	ACS (survey)	NES (tax)	NES (tax)
Geography	Comm. zone	State	State	State
Outcome	Incorp. SE rate	Incorp. SE rate	Corp. /1k	Corp. /1k
Denominator	Current	Current	Current	Baseline
Pre-trend $F(p)$	0.39 (0.81)	0.66 (0.63)	1.24 (0.31)	1.61 (0.19)
Unit $\times$ year FE	Yes	Yes	Yes	Yes
Sector $\times$ year FE	Yes	Yes	Yes	Yes

Notes: Two-phase difference-in-differences of the incorporation margin on standardized occupational AI exposure interacted with post-2015 indicators, reference years 2010–2014. Column 1 reproduces the main within-commuting-zone estimate. Columns 2–4 aggregate to state $\times$ sector, the finest geography at which the Nonemployer Statistics publish the legal-form split. Outcomes are the incorporated self-employment rate (columns 1–2, percentage points of employment) and the number of corporation nonemployer businesses per thousand workers (columns 3–4); the denominator is current-year employment in the same state-sector-year cell in every column except column 4, which divides by baseline (2005–2009) mean annual employment, a measure fixed before the estimation window opens. NES corporations combine the single pre-2014 “corporations” code with the C- and S-corporation codes used from 2014 onward. All columns weight by baseline cell size, cluster by the geographic unit, and saturate with unit-by-sector, unit-by-year, and sector-by-year fixed effects. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: AI exposure and the Section 199A service exclusion

Dependent margin	Incorporated (1)	Incorporated (2)	Incorporated (3)	Unincorporated (4)
AI exposure $\times$ 2015–2018	-0.144** (0.068)	-0.137* (0.071)		-0.130 (0.095)
AI exposure $\times$ 2019–2024	-0.401*** (0.088)	-0.407*** (0.095)		-0.071 (0.104)
SSTB share $\times$ 2015–2018		-0.022 (0.057)	-0.043 (0.055)	0.075 (0.061)
SSTB share $\times$ 2019–2024		0.017 (0.076)	-0.046 (0.070)	-0.023 (0.068)
Observations	165,495	165,495	165,495	165,495
SSTB share in specification	No	Yes	Only regressor	Yes
CZ $\times$ sector, CZ $\times$ year, sector $\times$ year FE	Yes	Yes	Yes	Yes

Notes: Two-phase difference-in-differences at the commuting-zone $\times$ sector $\times$ year level (2010–2024), reference years 2010–2014. AI exposure is the standardized Eloundou et al. (2024) measure; SSTB share is the standardized baseline (2005–2009) employment share of the thirty-nine occupations mapped to Section 199A specified-service fields, built by the shift-share construction of equation (1) on the same cells and the same workforce window as AI exposure. Column 1 is the main specification; column 2 adds SSTB share interacted with both phase indicators; column 3 includes only SSTB share; column 4 replaces the outcome with the unincorporated self-employment rate. The median-split estimates reported in the text come from a separate regression that splits cells at the employment-weighted within-sector median of SSTB share and includes SSTB-heavy $\times$ phase level controls. All columns include commuting-zone-by-sector, commuting-zone-by-year, and sector-by-year fixed effects, weight by baseline cell employment, and cluster standard errors by commuting zone. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

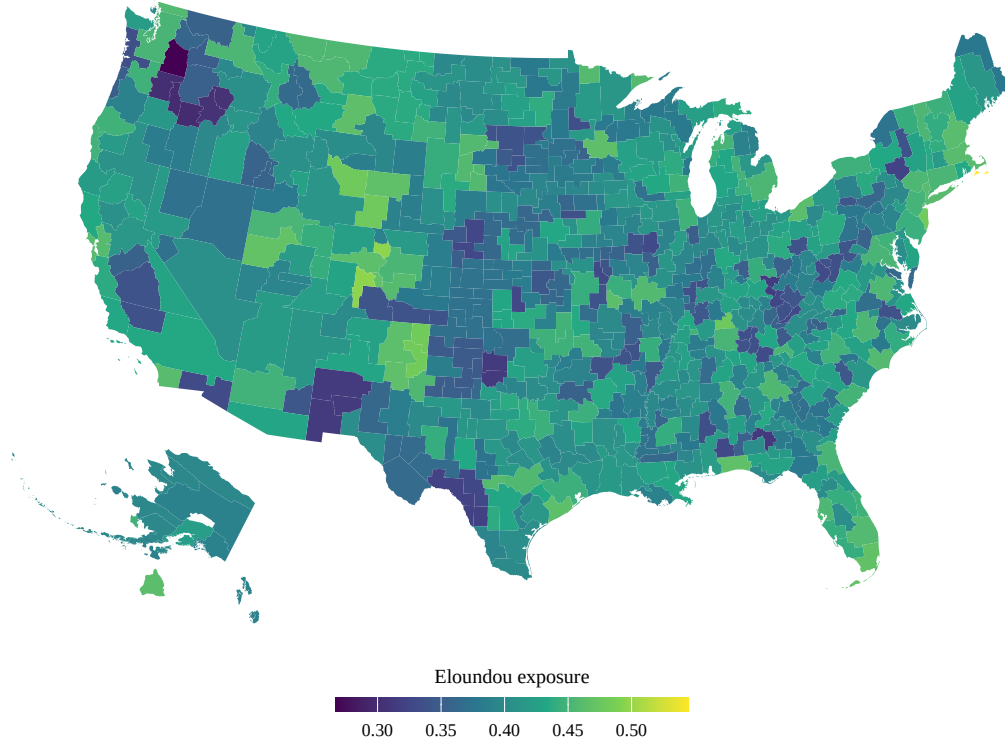


Figure 1: Occupational AI exposure in wholesale trade, by commuting zone

Notes: $AIExposure_{zk}$ from equation (1) for the wholesale trade sector in each of the 741 commuting zones, shown before standardization so the units are the raw Eloundou et al. (2024) occupation score. Counties are dissolved into commuting zones, so each shaded polygon is one commuting zone and no county borders are drawn. Alaska and Hawaii appear as insets. The occupation mix is the 2005–2009 ACS five-year file. Darker shading is higher exposure. Wholesale trade is shown because it is the third most AI-exposed of the fifteen sectors and one where the estimated decline is large and precise (Appendix Table C.10); the map is descriptive and involves no estimation.

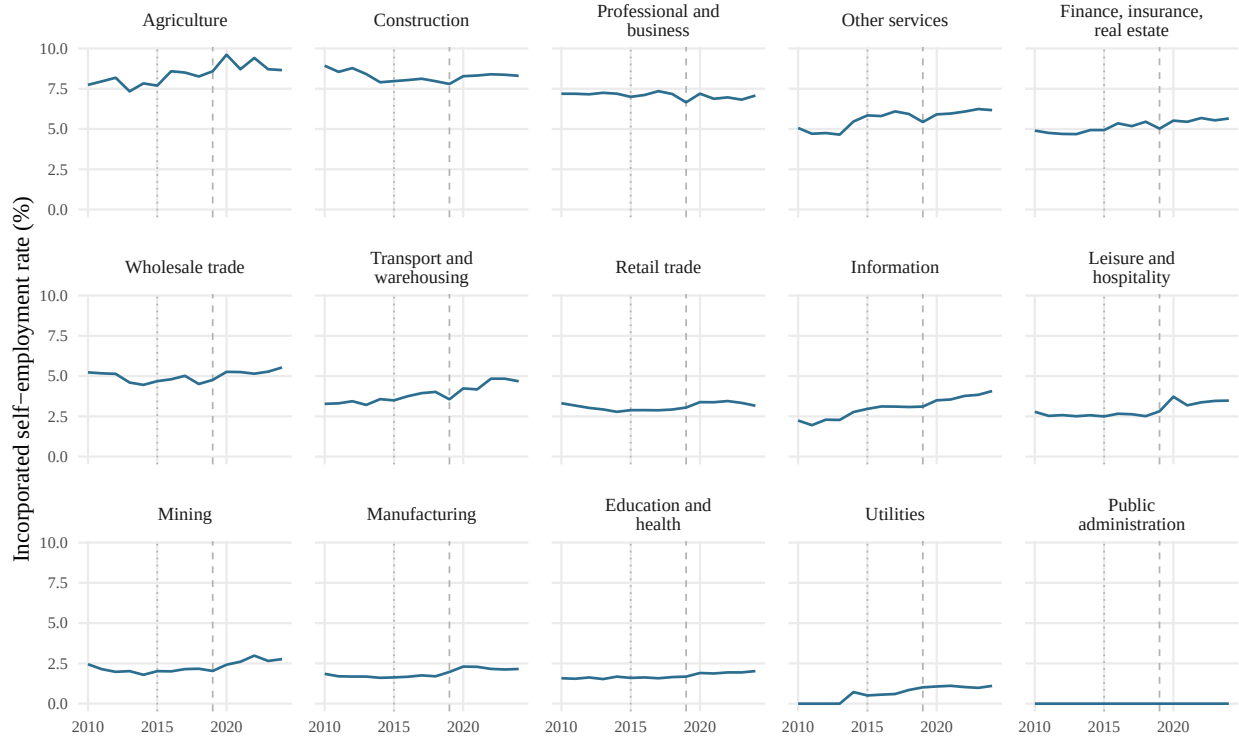


Figure 2: Incorporated self-employment by sector, 2010–2024

Notes: National incorporated self-employment rate, $100 \times \sum \text{se_inc} / \sum \text{emp}$, for each of the 15 Census major sectors, 2010–2024, from the ACS one-year files with person weights. The outcome is the incorporated margin only (CLASSWKRD 14), not total self-employment. Panels share a common vertical scale and are ordered by mean level. The dotted line marks the 2015 onset and the dashed line the 2019 diffusion boundary. Public administration is near zero throughout because incorporated self-employment is not an available legal form there. The 2020 ACS one-year file is the Census Bureau’s experimental release, reweighted for pandemic-era nonresponse; it is retained here and in the estimation panel, and the design’s commuting-zone-by-year and sector-by-year fixed effects absorb any level shift it induces.

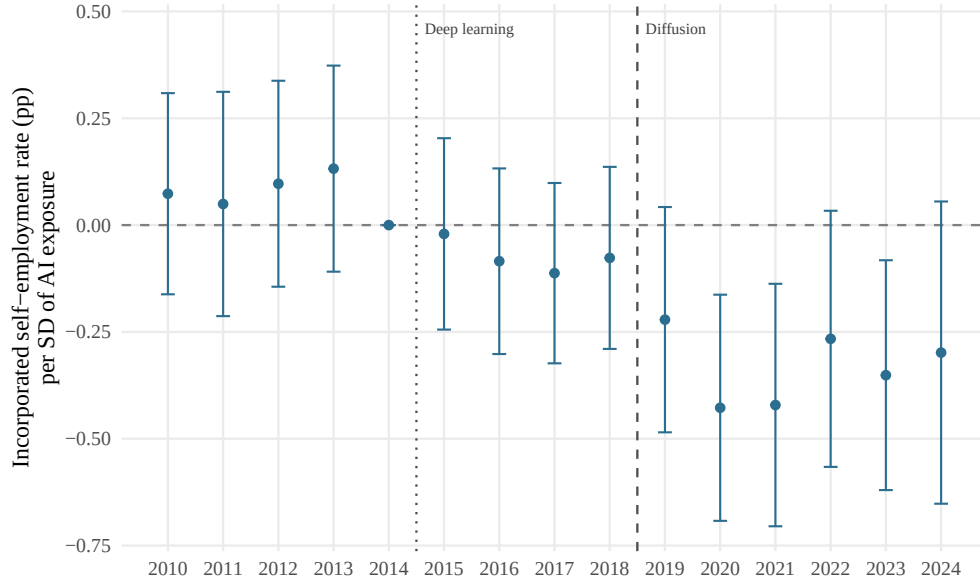


Figure 3: Incorporated self-employment in AI-exposed sectors: event study

Notes: Coefficients β_ℓ from equation (2) on $AIExposure_{zk} \times \mathbf{1}[t - T_0 = \ell]$, with $T_0 = 2015$ and reference period $\ell = -1$ (2014). Unit of observation: commuting zone $\times$ sector $\times$ year (741 commuting zones, 15 Census sectors, 2010–2024; 165,495 cells). Outcome: incorporated self-employment rate in percentage points (ACS CLASSWKRD 14). Exposure: standardized Eloundou et al. (2024) measure of equation (1). OLS with commuting-zone-by-sector, commuting-zone-by-year, and sector-by-year fixed effects, weighted by baseline (2005–2009) cell employment. Vertical bars are 95 percent confidence intervals from standard errors clustered by commuting zone. The dotted line marks the 2015 onset and the dashed line the 2019 diffusion boundary.

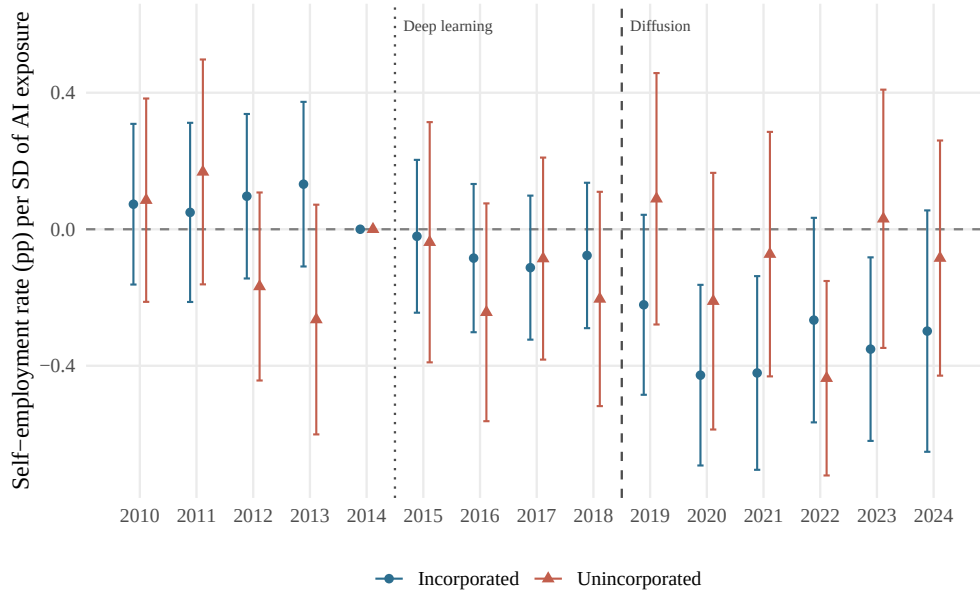


Figure 4: Incorporated versus unincorporated self-employment: event studies

Notes: Coefficients β_ℓ from equation (2) estimated separately for the incorporated (circles) and unincorporated (triangles) self-employment rates, both in percentage points per standard deviation of the Eloundou et al. (2024) exposure measure. Unit of observation: commuting zone $\times$ sector $\times$ year (741 commuting zones, 15 sectors, 2010–2024). OLS with commuting-zone-by-sector, commuting-zone-by-year, and sector-by-year fixed effects, weighted by baseline cell employment; reference period 2014. Vertical bars are 95 percent confidence intervals from standard errors clustered by commuting zone.

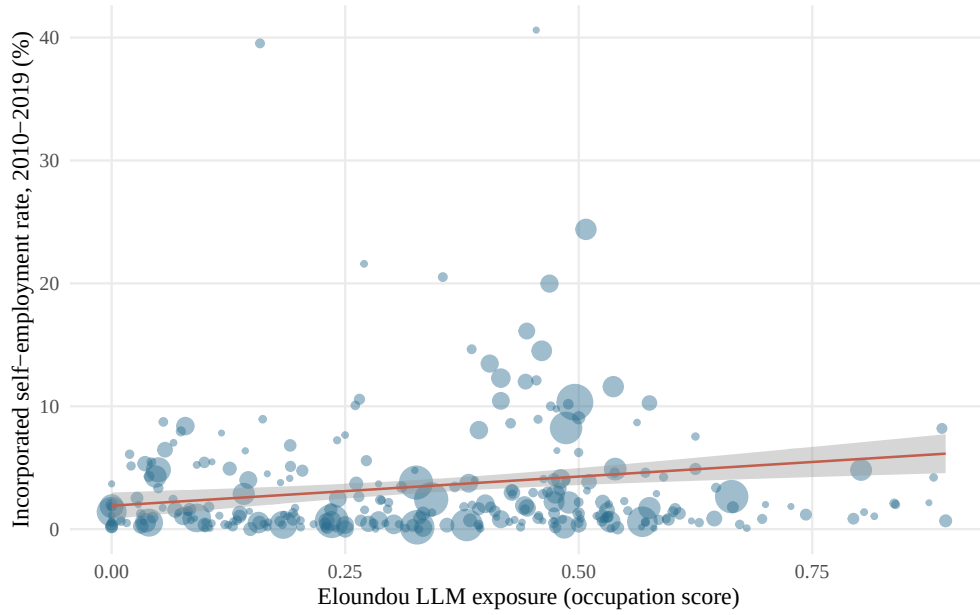


Figure 5: Incorporated self-employment and AI exposure across occupations

Notes: Unit of observation: OCC2010 occupation in the pooled 2010–2019 ACS (person-weighted). Horizontal axis: the occupation’s raw Eloundou et al. (2024) LLM exposure score; vertical axis: the occupation’s incorporated self-employment rate in percent (ACS CLASSWKRD 14). Point size is proportional to pooled employment; for readability the figure shows the 273 occupations with pooled employment above 500,000 (97 percent of employment). The line is the employment-weighted OLS fit with its 95 percent confidence band. Correlations reported in the text use all 408 occupations, employment-weighted. Descriptive cross-section; no fixed effects.

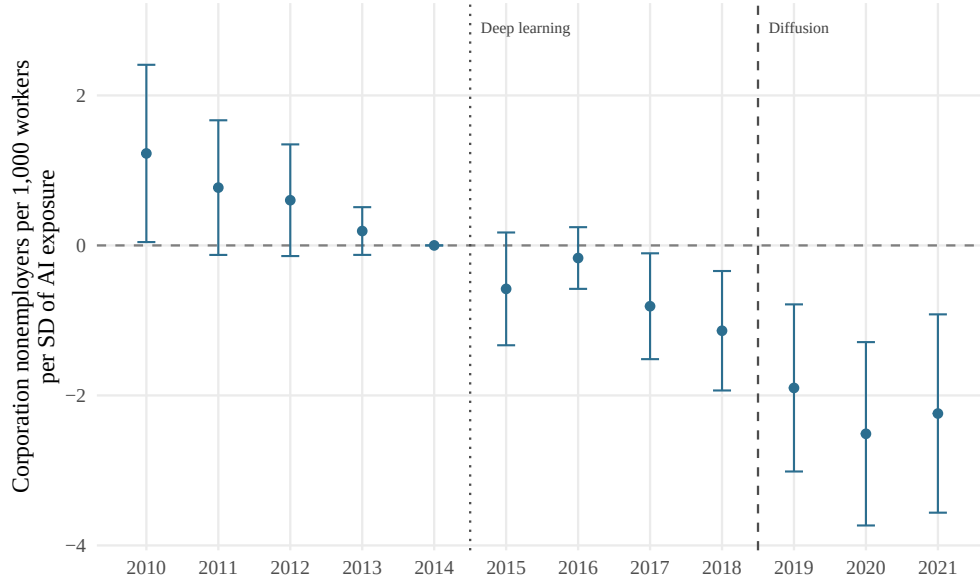


Figure 6: Corporation nonemployers in AI-exposed sectors: administrative-record event study

Notes: Event-study coefficients on standardized occupational AI exposure interacted with year indicators, onset year $T_0 = 2015$ and reference period 2014. Unit of observation: state $\times$ sector $\times$ year (51 states including the District of Columbia; 14 of the 15 Census major sectors, public administration having no nonemployer businesses; 2010–2021; 8,467 cells). Outcome: the number of corporation nonemployer businesses per thousand workers, the corporation count from the Census Nonemployer Statistics (IRS legal form of organization) divided by employment in the same state-sector-year cell from the ACS. Exposure: standardized Eloundou et al. (2024) measure. OLS with state-by-sector, state-by-year, and sector-by-year fixed effects, weighted by baseline cell employment. Vertical bars are 95 percent confidence intervals from standard errors clustered by state. The pre-2015 coefficients are jointly indistinguishable from zero ($F = 1.24$, $p = 0.31$).

A Conceptual Framework

This appendix gives the occupational-choice framework behind the paper’s estimates, built on a compact Lucas (1978) model of production.

A.1 AI as Capital

We infer that adopted AI is capital from Johnston and Makridis (2026), who find that wages rise and the labor share falls with AI exposure on the same Eloundou et al. (2024) measure this paper uses.

Take a constant-elasticity production function,

$$Y = [\alpha K^\rho + (1 - \alpha)L^\rho]^{1/\rho}, \quad \sigma = \frac{1}{1 - \rho}, \quad (\text{A1})$$

where Y is output, K adopted capital, L labor, $\alpha \in (0, 1)$ the capital weight, and $\rho < 1$ the substitution parameter. The elasticity σ measures how readily capital replaces labor, so the two are gross substitutes when $\sigma > 1$ and gross complements when $\sigma < 1$. Paying labor its marginal product, the wage and the labor share are

$$w = (1 - \alpha) \left(\frac{Y}{L} \right)^{1-\rho}, \quad \frac{wL}{Y} = (1 - \alpha) \left(\frac{L}{Y} \right)^\rho. \quad (\text{A2})$$

Neutral technical change scales both marginal products by the same factor, so the wage rises and the labor share does not move. A rise in the capital-labor ratio K/L also raises the wage, since it raises Y/L , but it moves the labor share as well, in a direction that depends on ρ . The share falls when $\rho > 0$, is unchanged in the Cobb-Douglas case $\rho = 0$, and rises when $\rho < 0$. A rising wage with a falling share therefore implies $\sigma > 1$.

Three consequences follow for the entry margin. A rising K/L raises the marginal product of labor, so the return to salaried work in exposed occupations rises. It raises the return to whoever else deploys the capital, so running a firm in those occupations becomes more productive as well. And operating at competitive scale now requires acquiring the capital, so the fixed cost of entering rises. The first and third make founding less attractive; the second makes it more so. Johnston and Makridis (2026) report the labor-share estimate as imprecise, so $\sigma > 1$ is an interpretation of their point estimate rather than a settled magnitude.

A.2 Occupational Choice

Workers are attached to occupation o , the same index the paper uses for the exposure score E_o and the baseline shares $s_{zko,0}$ of equation (1), and ability θ is distributed with CDF G_o . Let $A \geq 0$ denote adopted AI capital and $x_o \in [0, 1]$ the occupation's task-content exposure, the model counterpart of the measured score. Each worker chooses among salaried employment, incorporated entry (the growth-oriented margin of [Levine and Rubinstein 2017](#)), and own-account work, with payoffs

$$w_o(\theta, A) = \theta \omega_o(A), \quad v_o(\theta, A) = \theta p_o(A) - \varphi_o(A), \quad b_o(\theta), \quad (\text{A3})$$

where ω_o is the per-ability salaried return, p_o the per-ability return to running an incorporated firm, φ_o the setup and scale requirement, and b_o the own-account payoff. Exposure governs how adoption enters these payoffs. Where $x_o > 0$, the capital of Section A.1 raises the marginal product of whoever deploys it and has to be acquired before it can be deployed, so $\omega'_o(A) > 0$ and $p'_o(A) > 0$ alongside $\varphi'_o(A) > 0$. Adoption reaches none of the three where $x_o = 0$.

The return to ability is steepest in entrepreneurship and flattest in own-account work, $b'_o(\theta) < \omega_o(A) < p_o(A)$, over the range of adoption where entry remains viable. Within that range, workers sort by ability at two cutoffs, each the ability at which two of the payoffs in (A3) are equal. At θ_U own-account work and salaried employment pay the same, and at θ_E founding and salaried employment pay the same. Workers below θ_U take own-account work, those between the two take salaried employment, and those at or above θ_E found incorporated firms. Setting $v_o = w_o$ and solving for θ gives the entry cutoff and its response to adoption,

$$\theta_E(A) = \frac{\varphi_o(A)}{p_o(A) - \omega_o(A)}, \quad \frac{d\theta_E}{dA} = \underbrace{\frac{\varphi'_o}{p_o - \omega_o}}_{\text{entry cost rises}} - \underbrace{\frac{\varphi_o(p'_o - \omega'_o)}{(p_o - \omega_o)^2}}_{\text{founding advantage widens}}. \quad (\text{A4})$$

The cutoff is the ability at which the once-paid setup cost φ_o is just covered by the per-ability advantage $p_o - \omega_o$ of running a firm rather than working for one. A larger setup cost takes more ability to justify paying it, which pushes the cutoff up. A wider advantage lets less ability cover the same cost, which pulls it down. The advantage widens only if the owner's return gains more from adoption than the salaried wage does. Where the salaried wage gains more, $\omega'_o > p'_o$, it shrinks instead, both terms push the cutoff up, and far enough along entry closes altogether. Otherwise the two terms oppose each other, and incorporated ownership falls only when the setup cost grows proportionally faster than the advantage, $\varphi'_o/\varphi_o > (p'_o - \omega'_o)/(p_o - \omega_o)$. Where $x_o = 0$ all three derivatives vanish and the cutoff does

not move, so any response is confined to exposed occupations either way. The framework does not settle which side of the inequality holds. The decline this paper estimates is the evidence that the cost side dominates.

The own-account margin moves only a fraction of the entry margin, because own-account work sits in occupations with little exposure. Across the 408 occupations of Table C.4 the unincorporated rate correlates -0.17 with the Eloundou et al. (2024) score, and Panel B of Table 2 and Figure 4 show the muted response. The workers who stop entering are those just below the old cutoff, the top of the salaried ability distribution, hence established mid-career professionals rather than young first-time founders. That gradient in age and occupation appears in Tables C.11 and C.4 and Figure 5. Because θ_E depends on adopted capital rather than on the state of research, the break estimated in Figure 3 and Table C.5 is timed to diffusion (Babina et al., 2024), not to the earlier research breakthroughs.

The framework also states how this paper differs from the empirical AI-and-labor literature. That literature estimates how A moves ω_o and employment inside firms, the salaried branch of (A3) alone. Johnston and Makridis (2026) estimate the sector-level version, finding output, employment, and establishment counts all rising in exposed sectors. Neither their design nor the worker-level studies observe the v_o branch or the cutoff θ_E that separates it from w_o , because the Quarterly Census of Employment and Wages excludes the self-employed and payroll data cover only workers already inside firms. Our object is the cutoff. That is why the incorporated and unincorporated margins are estimated separately throughout, and why the paper’s identifying content lies in the contrast between them rather than in the level of either.

B Employee Wages in Exposed Cells

One candidate mechanism for the entry decline is that AI complements the scale, data, and organizational capital that incumbents hold, raising the salaried return to exposed cognitive work relative to the return from running an incorporated practice. This is the incumbent-wage channel, written as $\omega'_o(A) > 0$ in Appendix A. We measure the salaried side of that comparison directly. We regress the log person-weighted cell mean of wage and salary income among employees, meaning workers with positive wage income who are not incorporated owners, since the ACS records an incorporated owner’s self-paid salary as wage income, on standardized AI exposure interacted with the same two phase indicators, on the same panel, fixed effects, weights, and clustering as the main specification.

The diffusion-phase coefficient is negative. Cell mean employee wages fall by 0.014 log points per standard deviation of exposure in 2019–2024, significant at the 5 percent level, on

pre-trends that clear our threshold narrowly ($F = 2.16$, $p = 0.07$; column 1 of Table B.1).

That sign runs against Johnston and Makridis (2026), who estimate that a standard deviation of the same exposure measure raises hourly wages by 1.1 percent by 2024. Three differences in the estimand account for it. Their specification includes occupation fixed effects and therefore recovers a within-occupation price; dropping those fixed effects raises their own estimate to 5.6 percent, a gap between their columns wider than the gap between our two papers. Their sample is restricted to full-time workers aged 22 to 65 earning above \$31,200, while ours is an unconditional cell mean that retains part-time and marginal workers. And their outcome is an hourly wage where ours is annual wage and salary income. A cell mean moves with who is in the cell; a within-occupation hourly wage does not.

Our own estimates imply a composition shift of exactly that kind. The incorporated self-employment rate falls 0.401 percentage points per standard deviation while the unincorporated rate does not move, so the workers who do not become incorporated owners remain in the salaried pool. That pool holds roughly 90 percent of cell employment, which puts the flow at about 0.45 percent of it per standard deviation. A 0.45 percent increase in salaried labor supply against a 1.4 percent fall in the cell mean wage implies a labor demand elasticity near -0.3 . We report the arithmetic to show that our two coefficients are mutually consistent at an ordinary parameter value, not as an estimate of that elasticity, since the calculation assumes the deterred owners remain employed in the same cell and the cell is not a closed labor market. One compositional force runs against the observed sign rather than with it: the workers who do not enter incorporated ownership are drawn from the upper tail of the earnings distribution, so adding them to the salaried pool raises the measured mean, and any price decline must exceed 1.4 percent to produce the coefficient we estimate.

Columns 2 to 6 of Table B.1 report a companion exercise on the entrepreneurial side, the within-cell gap between log mean salaried earnings and log mean incorporated business income. We report it and do not build on it. The gap moves with the price of each option and with the re-sorting the entry decline itself produces, so cell means cannot separate the two, and the business-income measure is compromised at the source because S-corporation owners pay themselves salaries that the ACS records as wage income, leaving the gap defined in about 40 percent of cells. On the business-income construction of column 3 the diffusion-phase estimate is wrong-signed for a widening gap and its pre-2015 coefficients jointly reject parallel trends, so we draw no conclusion from it. On the total-own-income construction of column 5, which repairs the S-corporation artifact and covers 94 percent of baseline employment, the estimate is a precise null on flat pre-trends. The cell-level earnings data do not resolve the relative-payoff margin in either direction.

We therefore do not read this coefficient as a refutation of AI's effect on salaried pay, and

the entry result does not rest on it. A rising fixed cost of viable entry, $\varphi'_o(A) > 0$, deters founders whatever the salaried wage does, and Johnston and Makridis (2026) supply direct evidence for that reading on the employer side. They find that AI's output gains accrue disproportionately to capital, with about 29 percent of the induced output increase flowing to labor and the labor share falling by roughly 5 percentage points per standard deviation of exposure, an estimate they report as imprecise. Where production increasingly requires AI capital to remain competitive, the fixed cost of entry rises even as incumbent employment expands in the same sectors. Their finding that establishment counts rise 2.6 percent per standard deviation is consistent with our decline rather than in tension with it, because the Quarterly Census of Employment and Wages covers employer establishments and excludes the self-employed entirely.

Table B.1: Employee wages and the salaried–entrepreneurial earnings gap

Outcome	$\ln \bar{w}$ (1)	$\ln \hat{\pi}$ (2)	Gap (3)	Gap (4)	Gap, total inc. (5)	Gap, fixed age (6)
AI exposure $\times$ 2015–2018	-0.003 (0.005)	0.020 (0.155)	-0.024 (0.157)	-0.075 (0.172)	0.023 (0.023)	-0.020 (0.156)
AI exposure $\times$ 2019–2024	-0.014** (0.006)	0.232 (0.143)	-0.254* (0.144)	-0.299* (0.161)	0.006 (0.020)	-0.245* (0.145)
Observations	165,446	63,895	63,894	29,604	125,342	63,892
Sample	All	All	All	Emp $\geq$ 5k	All	All
Mean dep. var.	10.78	8.51	2.30	2.29	-0.54	2.30
CZ $\times$ sector FE	Yes	Yes	Yes	Yes	Yes	Yes
CZ $\times$ year FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector $\times$ year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Two-phase estimates of equation (3) for cell-level earnings outcomes at the commuting-zone $\times$ sector $\times$ year level (741 commuting zones, 15 Census sectors, 2010–2024), on cells where the column’s outcome is defined; the gap sample comprises 63,894 cells. Column 1: log person-weighted cell mean of wage and salary income (ACS INCWAGE) among employees (CLASSWKRD 20–28 with positive wage income); incorporated owners are excluded from the employee mean even though the ACS records their self-paid salaries as wage income. Column 2: log cell mean of business income (INCBUS00) among the incorporated self-employed (CLASSWKRD 14); negative and zero individual incomes remain in the mean, and the log is defined where the cell mean is positive. Column 3: the within-cell gap, column 1 minus column 2, where both are defined. Column 4 restricts to cells with baseline employment of at least 5,000. Column 5 rebuilds incorporated income as total own income, INCWAGE plus INCBUS00, because S-corporation owners pay themselves salaries reported as wage income. Column 6 holds each pool’s age composition fixed: means are computed within four age bins (16–29, 30–44, 45–59, 60 and over) and aggregated with the cell’s 2010–2014 bin employment shares, renormalized over non-empty bins. All columns include commuting-zone-by-sector, commuting-zone-by-year, and sector-by-year fixed effects, weight by baseline (2005–2009) cell employment, and cluster standard errors by commuting zone. Both income variables are top-coded and business losses are bottom-coded, which compresses the measured tails against the widening prediction. Mean dep. var. is the weighted sample mean of the outcome. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Additional Robustness Exhibits

This appendix collects the exhibits referenced in the text: the detailed-industry (IND1990) re-estimation and the occupational anatomy (Figure C.2 and Table C.4), the onset-timing split-stability panel (Table C.5), the Section 199A pre-recession-baseline horse race and its event-study companion (Table C.6 and Figure C.3), the occupation importance weights and leave-one-out (Tables C.7 and C.8), the share-balance and diffusion-window stability diagnostics (Table C.9), and the per-sector and age heterogeneity (Tables C.10 and C.11). The automating and augmentative decomposition of Section 4.3 is reported in Table C.3 and Figure C.1.

Table C.1: Occupational AI exposure and measured AI adoption

	AI use	Occupational exposure	
	(% of firms) (1)	Eloundou et al. (2024) (2)	Felten, Raj, and Seamans (2021) (3)
Information	6.74	+1.46	+1.53
Professional, business services	4.75	+0.89	+1.22
Finance, insurance, real estate	3.87	+1.87	+2.21
Education, health	3.24	+0.44	+0.91
Manufacturing	3.02	-0.12	-0.11
Utilities	2.81	+0.18	+0.32
Transportation, warehousing	2.52	-0.06	-0.73
Retail trade	2.38	+0.41	+0.05
Wholesale trade	2.25	+1.14	+0.82
Mining	1.93	-0.58	-0.38
Leisure, hospitality	1.93	-0.81	-0.76
Other services	1.79	-0.65	-0.21
Construction	1.71	-1.03	-1.25
Agriculture	1.47	-1.55	-1.12
Spearman rank correlation		+0.846	+0.868
Sectors		14	

Notes: Column 1 is the share of employer firms reporting any use of artificial intelligence in the 2018 Census Annual Business Survey, Technology and Business Practices module (code T1E03B99 as a share of T1E03BA9), aggregated from 2-digit NAICS to the fifteen Census sectors used throughout. Columns 2 and 3 are the employment-weighted sector means of the standardized exposure measures of equation (1). Public administration is absent from the survey frame, leaving fourteen sectors. Spearman rank correlations are significant at the 1 percent level ($p = 0.0001$ for both measures); the Pearson correlation with the Eloundou et al. (2024) measure is +0.745 ($p = 0.002$). The exercise validates the ordering of sectors only. The survey's state-by-sector tabulation suppresses 538 of 1,071 cells, and suppression concentrates in small, low-exposure sectors, so it cannot be used to validate the within-sector variation across commuting zones that identifies the estimates.

Table C.2: Exposure-robust inference for the shift-share design

	Standard error (1)	p -value (2)	95% confidence interval (3)
Clustered by commuting zone (baseline)	0.088	< 0.001	$[-0.577, -0.233]$
Heteroskedasticity-robust	0.078	< 0.001	$[-0.558, -0.252]$
Exposure-robust (AKM)	0.068	< 0.001	$[-0.539, -0.271]$
Exposure-robust, null-imposed (AKM0)	0.071	< 0.001	$[-0.543, -0.267]$
Point estimate		-0.405	
Observations		11,109	

Notes: The Adão, Kolesár, and Morales (2019) correction is defined for a shift-share cross-section, so the panel is collapsed to the long difference the two-phase specification identifies: the 2019–2024 cell mean of the incorporated self-employment rate minus the 2010–2014 mean, regressed on standardized Eloundou et al. (2024) exposure with commuting-zone and sector fixed effects, weighted by baseline (2005–2009) cell employment. The collapsed point estimate is -0.405 percentage points per standard deviation, against -0.338 in the corresponding panel specification. Shares are the 2005–2009 occupation employment shares within each commuting-zone-by-sector cell across 450 occupations, the same object equation (1) aggregates into the treatment. AKM0 imposes the null when constructing the interval. All intervals use the collapsed point estimate.

Table C.3: Automating and augmentative AI exposure

Outcome	Incorporated (1)	Unincorporated (2)
Automating exposure $\times$ 2015–2018	-0.279*** (0.075)	0.024 (0.103)
Augmentative exposure $\times$ 2015–2018	0.075 (0.065)	-0.130 (0.101)
Automating exposure $\times$ 2019–2024	-0.164* (0.087)	0.244** (0.120)
Augmentative exposure $\times$ 2019–2024	-0.283*** (0.070)	-0.275** (0.113)
Automating: diffusion minus onset <i>p</i> -value	+0.115 (0.081) 0.155	+0.219 (0.089) 0.013
Augmentative: diffusion minus onset <i>p</i> -value	-0.357 (0.080) <0.001	-0.145 (0.072) 0.044
Mean dep. var.	3.71	6.05
CZ $\times$ sector FE	Yes	Yes
CZ $\times$ year FE	Yes	Yes
Sector $\times$ year FE	Yes	Yes
Observations	165,495	165,495

Notes: Two-phase estimates of equation (3) with AI exposure split into its automating and augmentative components, both entered jointly. Unit of observation: commuting zone $\times$ sector $\times$ year (741 commuting zones, 15 Census sectors, 2010–2024). Automating exposure aggregates the Eloundou et al. (2024) E1 share, the tasks a large language model completes at least twice as fast on its own; augmentative exposure aggregates the E2 share, the tasks requiring complementary software or human collaboration to achieve the same speedup. Both are built from 2005–2009 baseline occupation shares using the score-matched denominator of the published measure, and each is standardized by its own cell standard deviation, so the two coefficients are not shares of one total and do not sum to the headline estimate. Outcomes are self-employment rates in percentage points. Both columns include commuting-zone-by-sector, commuting-zone-by-year, and sector-by-year fixed effects, weight by baseline (2005–2009) cell employment, and cluster standard errors by commuting zone. The diffusion-minus-onset rows test equality of the two window coefficients for each component. The two components correlate 0.58 across cells (0.35 after commuting-zone and sector fixed effects, variance inflation factor 1.14), so the decomposition is descriptive and does not identify a separate channel. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.4: Occupational anatomy of incorporated self-employment

<i>Panel A: Employment-weighted correlations of the occupational self-employment rate with AI exposure (408 occupations)</i>		
	Eloundou	Felten
Incorporated rate	0.20	0.23
Unincorporated rate	-0.17	-0.15
<i>Panel B: Exposure-tercile decomposition (occupations ranked into equal-count terciles by Eloundou exposure)</i>		
	Share of incorp. SE (%)	Share of employment (%)
Top tercile	56.0	37.7
Middle tercile	26.1	37.5
Bottom tercile	17.9	24.8
<i>Panel C: Illustrative occupations</i>		
	Incorp. rate (%)	Exposure percentile
Chief executives	24	79
Real-estate brokers	20	71
Lawyers	15	70

Notes: Occupation-level cross-section from the pooled 2010–2019 ACS across the 408 OCC2010 occupations, employment-weighted throughout. Panel A reports employment-weighted Pearson correlations between an occupation’s self-employment rate (incorporated or unincorporated, as a share of the occupation’s employment) and its exposure score; the Felten correlations exclude the five occupations lacking an AIOE score. Panel B ranks occupations into equal-count terciles by Eloundou et al. (2024) exposure and reports each tercile’s share of aggregate incorporated self-employment and of employment. Panel C gives the incorporated self-employment rate and the exposure-rank percentile (unweighted, among the 408 occupations) for three illustrative occupations. Exposure scores are the raw Eloundou et al. (2024) (Eloundou) and Felten, Raj, and Seamans (2021) (Felten) measures. The composition is descriptive and includes no fixed effects.

Table C.5: Onset timing: split-stability of the diffusion window

Diffusion window begins	2018 (1)	2019 (2)	2020 (3)
AI exposure $\times$ onset window	-0.143* (0.074)	-0.144** (0.068)	-0.173** (0.069)
AI exposure $\times$ diffusion window	-0.365*** (0.080)	-0.401*** (0.088)	-0.424*** (0.089)
Observations	165,495	165,495	165,495
CZ $\times$ sector FE	Yes	Yes	Yes
CZ $\times$ year FE	Yes	Yes	Yes
Sector $\times$ year FE	Yes	Yes	Yes

Notes: Two-phase difference-in-differences of the incorporated self-employment rate on standardized Eloundou et al. (2024) AI exposure at the commuting-zone $\times$ sector $\times$ year level (2010–2024). Columns differ only in the year the diffusion window begins; the onset window runs from 2015 to the year before. All columns include commuting-zone-by-sector, commuting-zone-by-year, and sector-by-year fixed effects, weight by baseline (2005–2009) cell employment, and cluster by commuting zone. A least-squares structural break on the event-study coefficient path (searched over onset years 2016–2021, with a 4,000-draw parametric bootstrap) falls at 2019, with a 95 percent confidence interval of [2016, 2020] that reflects the mild but already-negative onset-window coefficients preceding the sharper 2019 break; the sup-Wald statistic is 23.5 against a 5 percent critical value of 8.85, and the break-date distribution places probability 0.04 on 2018. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.6: Section 199A horse race on the pre-recession baseline

	(1)	(2)
AI exposure $\times$ 2015–2018	-0.137* (0.071)	-0.084 (0.070)
AI exposure $\times$ 2019–2024	-0.407*** (0.095)	-0.326*** (0.089)
SSTB share $\times$ 2015–2018	-0.022 (0.057)	0.004 (0.054)
SSTB share $\times$ 2019–2024	0.017 (0.076)	0.008 (0.069)
Exposure baseline window	2005–2009	2005–2007
corr(SSTB share, AI exposure)	0.517	0.506
Observations	165,495	165,425
CZ $\times$ sector FE	Yes	Yes
CZ $\times$ year FE	Yes	Yes
Sector $\times$ year FE	Yes	Yes

Notes: The Section 199A horse race of Table 6 re-run with both AI exposure and specified-service (SSTB) share built on the pre-recession 2005–2007 window rather than 2005–2009, from the 1-year ACS files with the vintage-matched PUMA-to-commuting-zone crosswalk. Column 1 reproduces the 2005–2009 horse race; column 2 uses the 2005–2007 baseline for both measures. Outcome: incorporated self-employment rate. Both columns include commuting-zone-by-sector, commuting-zone-by-year, and sector-by-year fixed effects, weight by baseline cell employment, and cluster by commuting zone. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.7: Occupation importance weights

Occupation	Eloundou	Mean share (%)	Importance
Customer Service Representatives	0.57	1.81	0.076
Tellers	0.54	0.66	0.068
Secretaries and Admin. Assistants	0.66	3.46	0.068
Insurance Sales Agents	0.58	0.58	0.064
Supervisors, Non-Retail Sales	0.54	1.56	0.062
Sales Reps., Wholesale/Manufacturing	0.54	1.57	0.057
Supervisors, Construction/Extraction	0.42	1.45	-0.054
Real Estate Brokers and Sales Agents	0.47	0.63	0.053
Financial Managers	0.47	0.76	0.048
Misc. agricultural workers	0.08	3.85	-0.043
Credit Counselors and Loan Officers	0.61	0.36	0.040
Supervisors, Office/Admin. Support	0.49	1.27	0.037
Bookkeeping/Accounting Clerks	0.80	1.54	0.036
Misc. managers	0.50	2.73	0.034
Property/Real Estate Managers	0.44	0.43	0.030

Notes: Rotemberg-style importance weights for the fifteen occupations contributing most to the variance of AI exposure across commuting-zone $\times$ sector cells, $\alpha_o = E_o \text{Cov}(s_{zko,0}, AIE_{\text{exposure}_{zk}}) / \text{Var}(AIE_{\text{exposure}_{zk}})$, which sum to one across all occupations. Eloundou is the occupation's raw exposure score and mean share its employment-weighted baseline (2005–2009) share. The ten highest-weight occupations account for 34 percent of the total absolute weight.

Table C.8: Occupation leave-one-out

Exposure measure	AI $\times$ 2019–2024	p
Baseline (all occupations)	-0.401*** (0.088)	0.000
<i>Drop each top-importance occupation</i>		
Customer Service Representatives	-0.411*** (0.085)	0.000
Tellers	-0.400*** (0.087)	0.000
Secretaries and Admin. Assistants	-0.394*** (0.089)	0.000
Insurance Sales Agents	-0.400*** (0.087)	0.000
Supervisors, Non-Retail Sales	-0.397*** (0.087)	0.000
<i>Drop cumulatively</i>		
Top 5	-0.393*** (0.083)	0.000
Top 10	-0.420*** (0.080)	0.000

Notes: The 2019–2024 two-phase coefficient when the AI-exposure measure is rebuilt excluding the highest-importance occupations of Table C.7, dropped one at a time and cumulatively, and re-standardized after each exclusion. The baseline row uses all occupations and reproduces the headline estimate. All specifications include commuting-zone-by-sector, commuting-zone-by-year, and sector-by-year fixed effects, weight by baseline cell employment, and cluster by commuting zone. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.9: Design diagnostics: share balance and diffusion-window stability

	Statistic	p
<i>Panel A: Pre-2015 outcome trends on top-10 occupation shares</i>		
Joint test, no fixed effects	$F = 12.63$	0.000
Joint test, within CZ and sector	$F = 0.73$	0.697
<i>Panel B: Stability of the 2019–2024 coefficients</i>		
Equality of the 6 coefficients	$\chi^2_5 = 3.23$	0.664
Linear trend, pp per year	-0.014 (0.025)	0.580

Notes: Panel A regresses each commuting-zone $\times$ sector cell’s pre-2015 linear trend in the incorporated self-employment rate (2010–2014, cells with at least three years) on the baseline (2005–2009) employment shares of the ten highest-importance occupations of Table C.7, weighted by baseline cell employment with standard errors clustered by commuting zone; reported are joint F -tests that all ten share coefficients are zero, without fixed effects and absorbing commuting-zone and sector fixed effects (the two-way comparison the estimator makes). Panel B tests the stability of the five 2019–2024 event-study coefficients of equation (2): a Wald test of their equality using the cluster-robust covariance, and the GLS slope of the coefficients on years since 2019 with its standard error in parentheses.

Table C.10: Per-sector estimates

	AI $\times$ 2015–2018	AI $\times$ 2019–2024
Agriculture	0.007 (0.569)	-0.319 (0.434)
Construction	-0.609 (0.596)	-2.082*** (0.478)
Professional & business	-0.169 (0.251)	-0.515** (0.223)
Other services	-0.498** (0.227)	-0.852*** (0.318)
Finance, insurance, real estate	-1.076** (0.459)	-0.572 (0.585)
Wholesale trade	0.054 (0.232)	-0.937*** (0.330)
Transport & warehousing	0.191 (0.455)	-0.066 (0.348)
Retail trade	-0.206 (0.363)	-0.319 (0.406)
Information	-0.002 (0.381)	-0.542* (0.321)
Leisure & hospitality	-0.262 (0.401)	-0.094 (0.347)
Mining	-0.186 (0.121)	-0.139 (0.212)
Manufacturing	-0.007 (0.069)	-0.227** (0.089)
Education & health	-0.104 (0.213)	-0.406* (0.231)
Utilities	-0.199 (0.126)	-0.264* (0.135)
Public administration	-0.168*** (0.053)	0.009 (0.078)
Observations	165,495	
CZ $\times$ sector, CZ $\times$ year, sector $\times$ year FE	Yes	

Notes: Sector-specific coefficients from a single regression on the full panel that interacts standardized AI exposure $\times$ phase with sector indicators, retaining commuting-zone-by-sector, commuting-zone-by-year, and sector-by-year fixed effects; each sector's coefficient is identified from cross-commuting-zone variation in exposure within that sector, an estimand that complements rather than reproduces the pooled coefficient. Sectors are ordered by baseline incorporated self-employment intensity. Weighted by baseline cell employment, standard errors clustered by commuting zone. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.11: Heterogeneity by age

Age group	16–29 (1)	30–44 (2)	45–59 (3)	60+ (4)
AI exposure $\times$ 2015–2018	-0.009 (0.088)	-0.184** (0.091)	-0.134 (0.109)	-0.171 (0.213)
AI exposure $\times$ 2019–2024	-0.168* (0.100)	-0.382*** (0.107)	-0.398*** (0.141)	-0.436** (0.213)
Mean incorporated-SE rate (%)	1.10	3.22	4.81	6.80
Observations	159,596	163,523	164,023	160,747

Notes: Two-phase estimates of the incorporated self-employment response within four age bins, each from the main specification re-estimated on a commuting-zone $\times$ sector $\times$ year panel built for that bin and weighted by the bin's baseline (2005–2009) employment. All include commuting-zone-by-sector, commuting-zone-by-year, and sector-by-year fixed effects and cluster by commuting zone. The mean incorporated self-employment rate is reported per bin. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

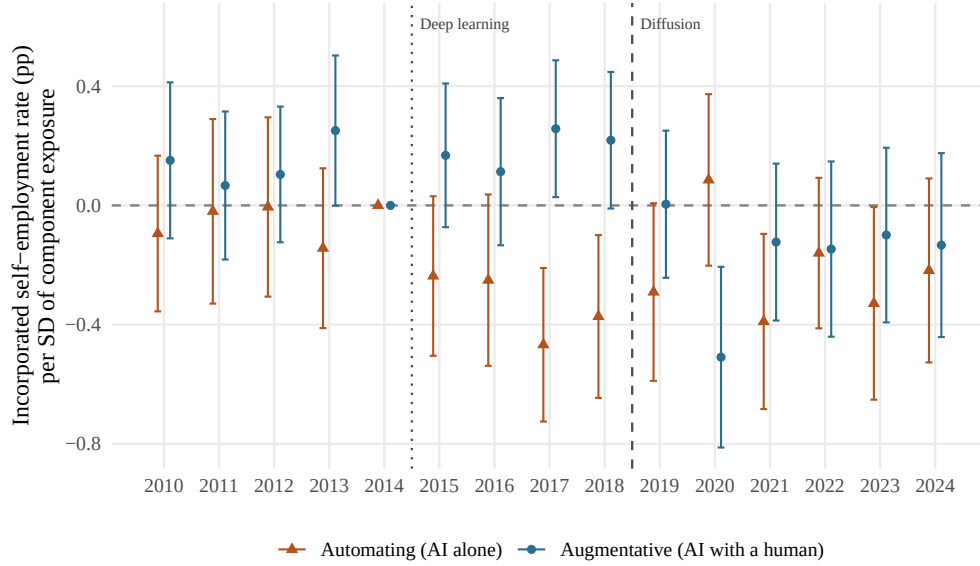


Figure C.1: Automating versus augmentative exposure: event studies

Notes: Coefficients from equation (2) re-estimated with the automating and augmentative components of AI exposure entered jointly, reference period 2014. Outcome: incorporated self-employment rate in percentage points. All specifications include commuting-zone-by-sector, commuting-zone-by-year, and sector-by-year fixed effects, weight by baseline (2005–2009) cell employment, and cluster standard errors by commuting zone. Vertical bars are 95 percent confidence intervals. The dotted line marks the 2015 onset and the dashed line the 2019 diffusion boundary. The joint test that the pre-2015 coefficients are zero does not reject for either component: $\chi^2(4) = 1.80$ ($p = 0.77$) for automating exposure and 4.09 (0.39) for augmentative exposure. The augmentative coefficient is -0.51 in 2020 and between -0.10 and -0.15 in each year from 2021 to 2024, so the pooled diffusion-window estimate of Table C.3 is weighted toward 2020.

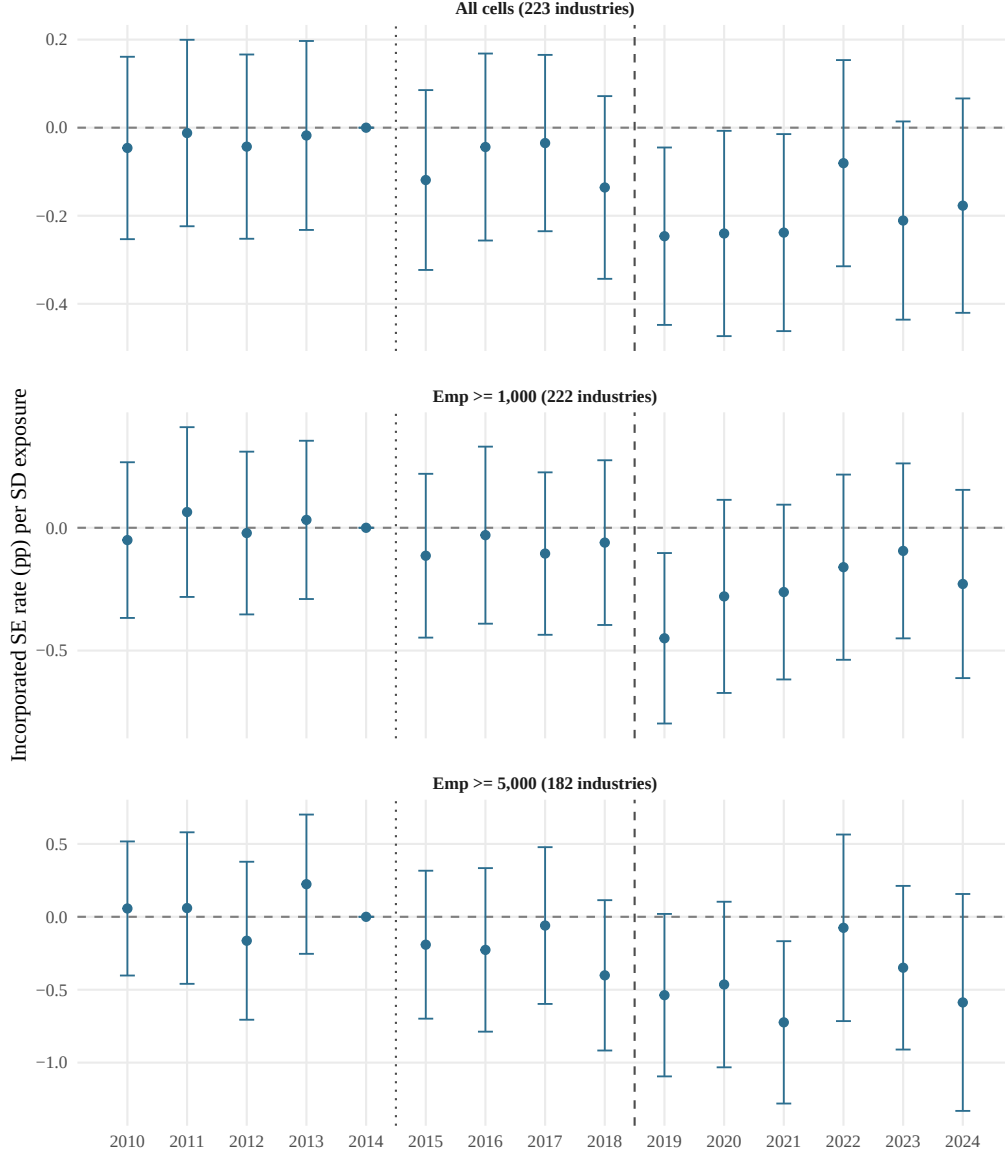


Figure C.2: Detailed-industry (IND1990) event studies

Notes: Coefficients β_ℓ from equation (2) re-estimated with the 223 harmonized detailed industries of the IPUMS IND1990 classification replacing the 15 Census major sectors, at the commuting-zone $\times$ industry $\times$ year level (2010–2024). Outcome: incorporated self-employment rate in percentage points; exposure is the standardized Eloundou et al. (2024) measure, reference year 2014 (plotted at zero). The top panel uses all cells; the lower panels restrict to cells with baseline employment of at least 1,000 and 5,000 workers, and the industry count in each panel header is the number surviving that floor. All panels include commuting-zone-by-industry, commuting-zone-by-year, and industry-by-year fixed effects, weight by baseline (2005–2009) cell employment, and cluster standard errors by commuting zone. Vertical bars are 95 percent confidence intervals. The dotted line marks the 2015 onset and the dashed line the 2019 diffusion boundary. The joint test that the pre-2015 coefficients are zero does not reject in any panel: $F = 0.08$ ($p = 0.99$) using all cells, 0.15 (0.96) at the 1,000-worker floor, and 0.57 (0.69) at the 5,000-worker floor. Panels carry independent vertical scales because precision falls sharply as the employment floor rises.

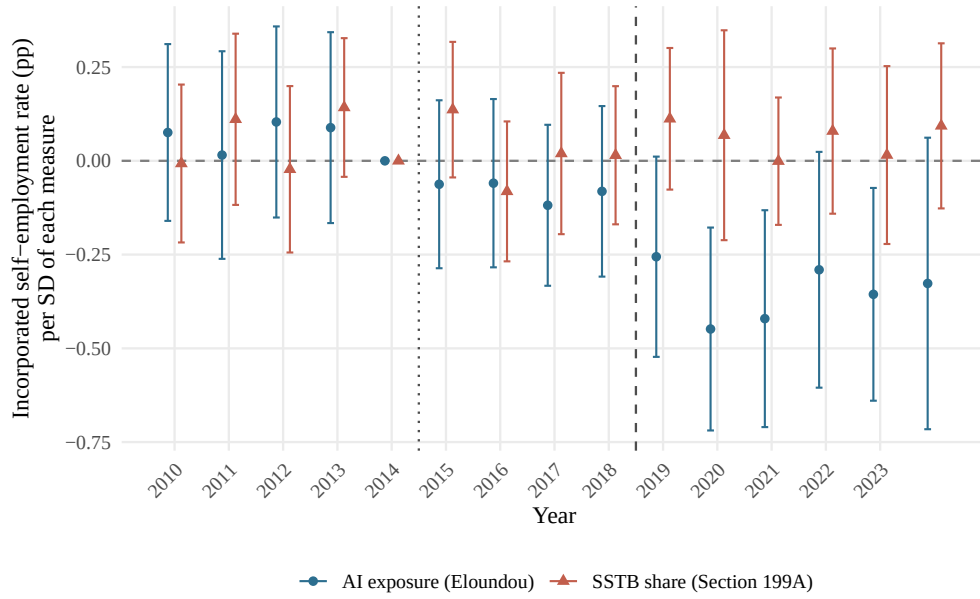


Figure C.3: AI exposure and Section 199A service share: event-study horse race

Notes: Event-study coefficients on standardized AI exposure (circles) and standardized Section 199A service share (triangles), each interacted with year indicators, onset $T_0 = 2015$ and reference period 2014, at the commuting-zone $\times$ sector $\times$ year level (2010–2024). Both measures enter the same regression. Outcome: incorporated self-employment rate in percentage points. The dotted line marks the 2015 onset and the dashed line the 2019 diffusion break; the first Tax Cuts and Jobs Act tax year is 2018. Commuting-zone-by-sector, commuting-zone-by-year, and sector-by-year fixed effects, weighted by baseline cell employment, standard errors clustered by commuting zone. Bars are 95 percent confidence intervals.